

Intro to Computing at CARC with Crystallography Example

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Version 0.1



Topics

- 1) Linux BASH Shell
- 2) The purpose of High-Performance Computing (HPC)
- 3) Running Programs on an HPC system (Slurm)
- 4) Analyse **Thaumatococcus**

NOTE: All these topics are covered in our online video tutorials

Search bar: Type to search



Logging into Hopper



```
ssh vanilla@hopper.alliance.unm.edu
```

You may be prompted for a password...

Don't let me move on until you are able to login.

Logging into Hopper

Welcome to Hopper

Be sure to review the "Acceptable Use" guidelines posted on the CARC website.

For assistance using this system email help@carc.unm.edu.

Tutorial videos can be accessed through the CARC website: Go to <http://carc.unm.edu>, select the "New Users" menu and then click "Introduction to Computing at CARC".

Warning: By default home directories are world readable. Use the `chmod` command to restrict access.

Don't forget to acknowledge CARC in publications, dissertations, theses and presentations that use CARC computational resources:

"We would like to thank the UNM Center for Advanced Research Computing, supported in part by the National Science Foundation, for providing the research computing resources used in this work."

Please send citations to publications@carc.unm.edu.

Hopper is our newest general purpose cluster and member of the Taos/Hopper condo.

There are four types of slurm partitions on Hopper:

- 1) General - this partition is accessible by all CARC users.
- 2) Debug - for testing your code and interactive jobs. Short time limits so that nodes are usually available right away.
- 3) Condo - this partition is accessible by users who are members of the Hopper/Taos condo. Users with access to this partition also have access to the community partition on the Taos cluster. Jobs run here may be interrupted by the hardware owner, so we recommend that your software support checkpointing so it can recover.
- 4) Private partitions - these partitions are dedicated to the condo grant/lab/center that purchased the associated hardware.

Type "qgrok" to get information about the partitions to which you have access.

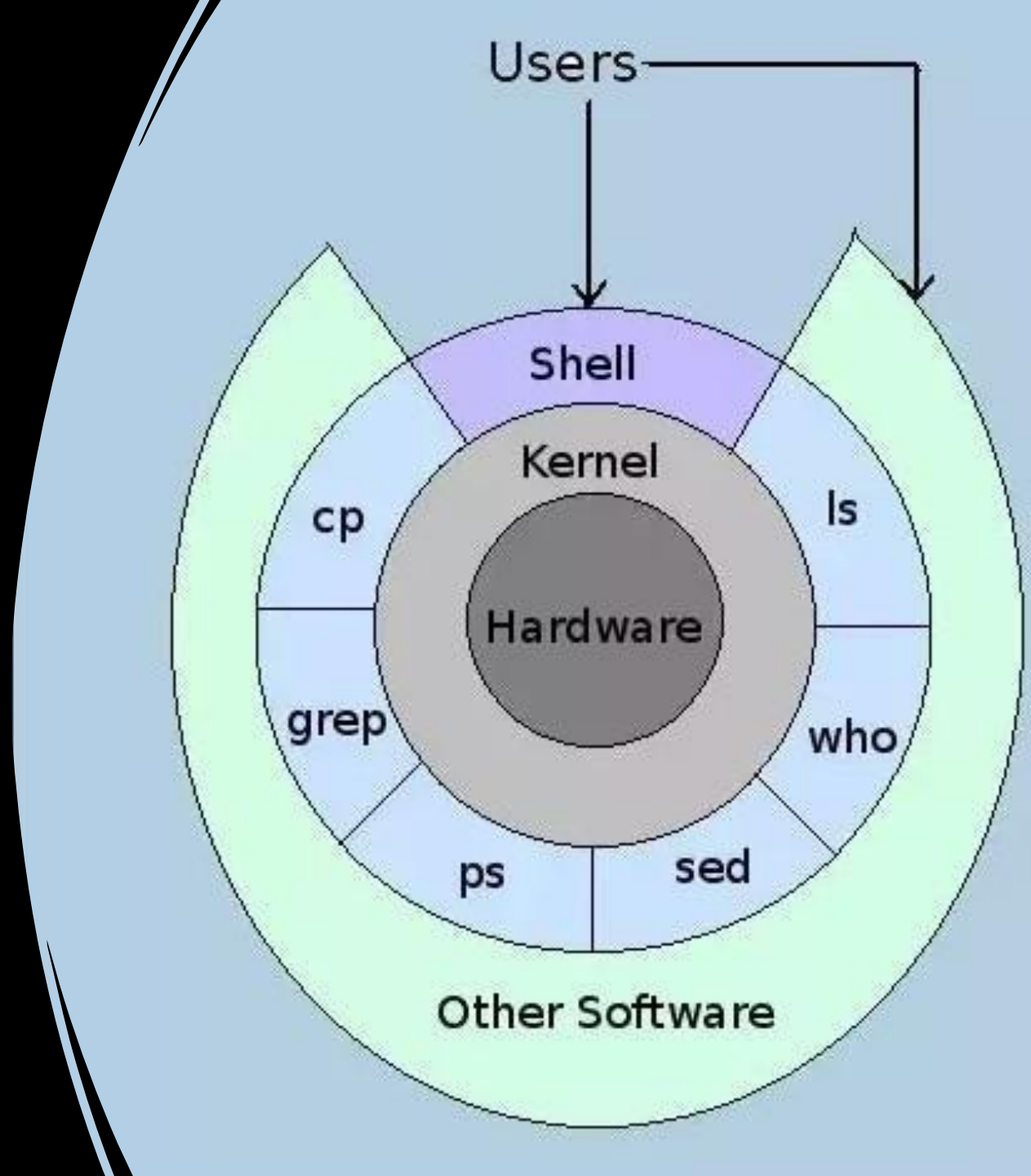
Enter "quotas" to see your storage usage and limits.

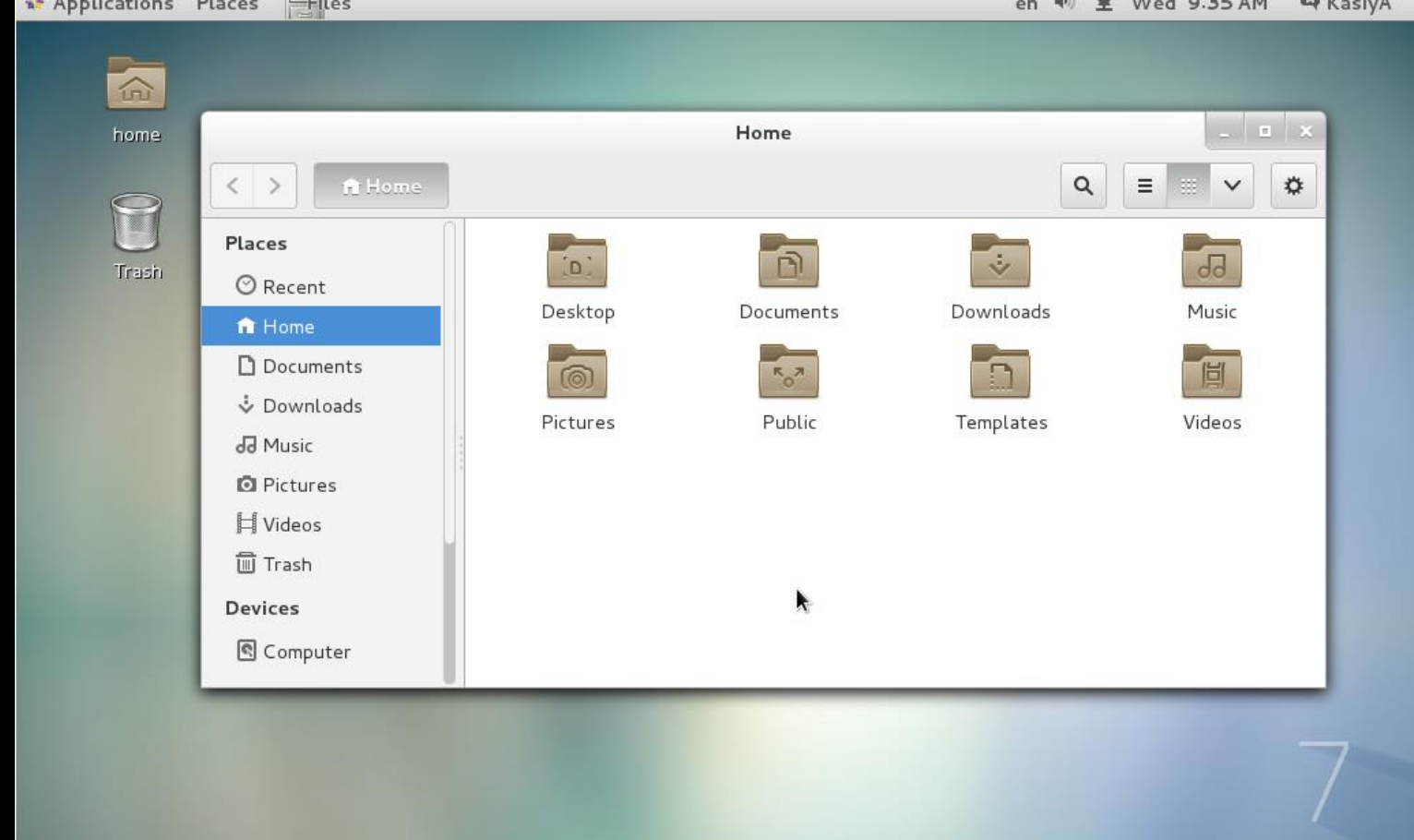
For a list of software installed on Hopper enter "module spider".

```
*****
* Logging in under another person's account is strictly forbidden and will *
* result in the account being          locked.                             *
*****
```

Linux and the BASH Shell

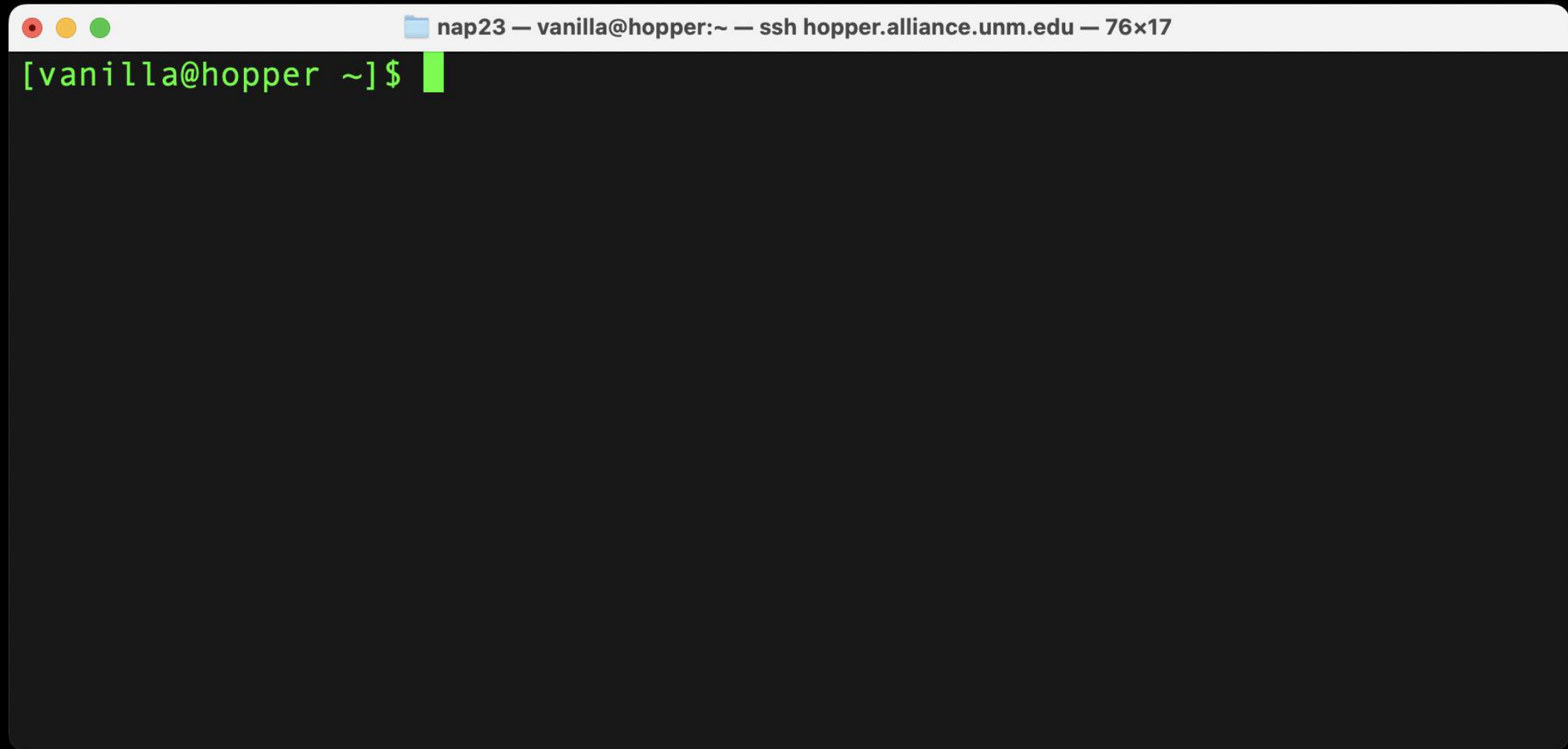
- The Kernel manages access to the hardware in a computer.
- An Operating System (OS) is the Kernel plus useful programs provided by the OS.
- The “shell” is the outermost layer of the OS.
- It is where the user interacts with the OS.





Graphical Shells (GUIs)

Logging into Hopper





**YOU ARE AT THE
RIGHT PLACE**

Linux and the BASH Shell

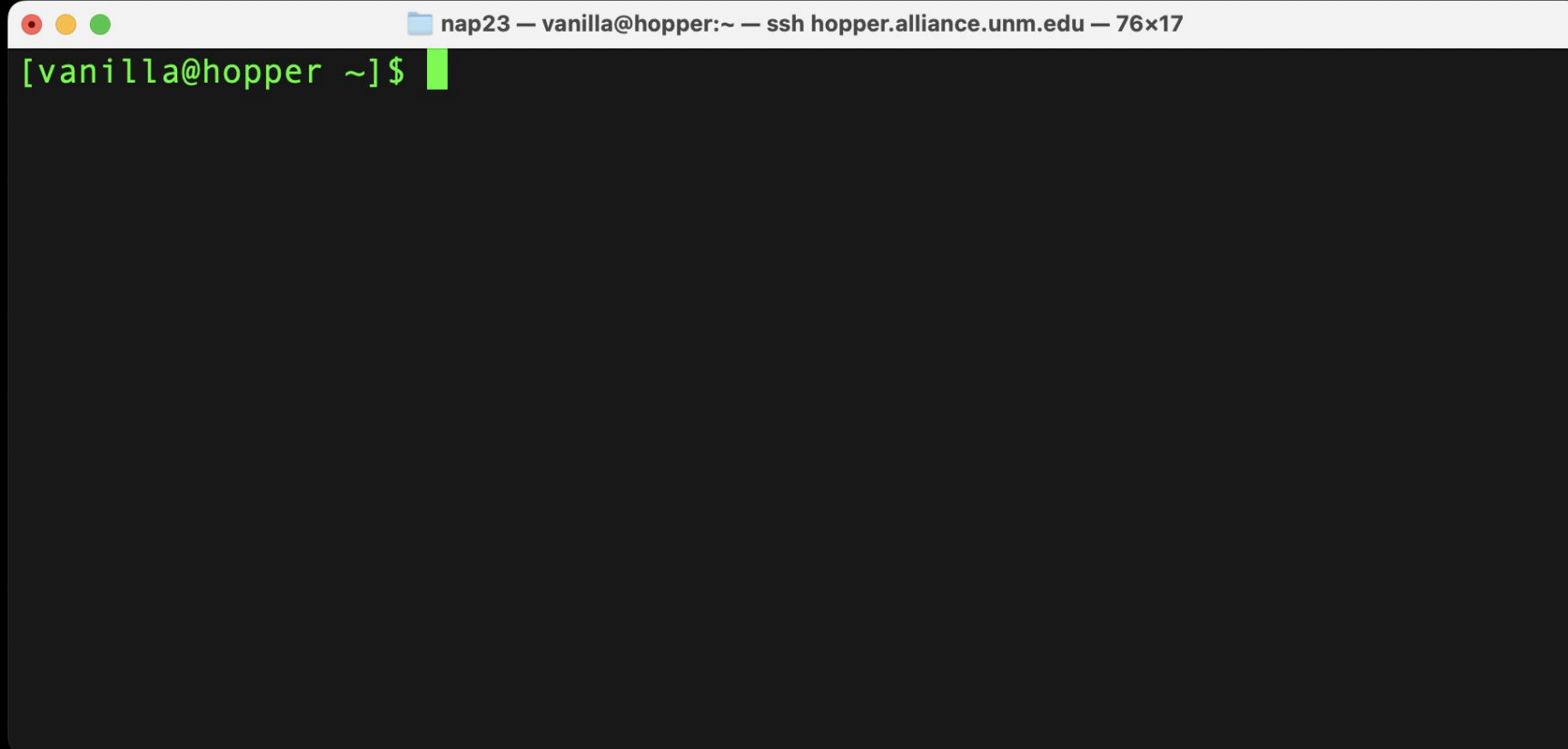


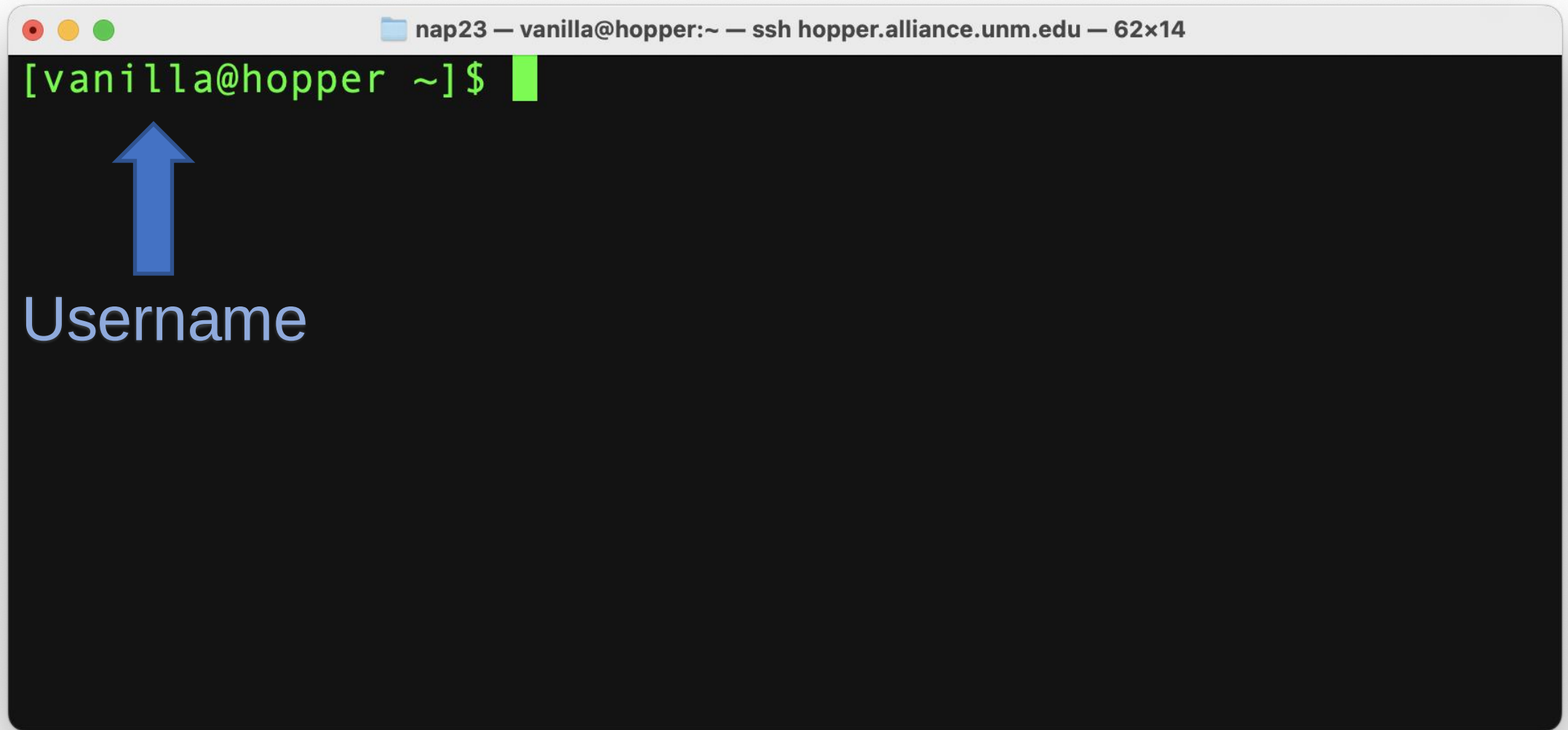
Linux

WHERE THERE IS A SHELL, THERE IS A WAY!

The Borne-Again Shell (BASH)

Written in 1976 by Stephen Bourne for UNIX version 7.



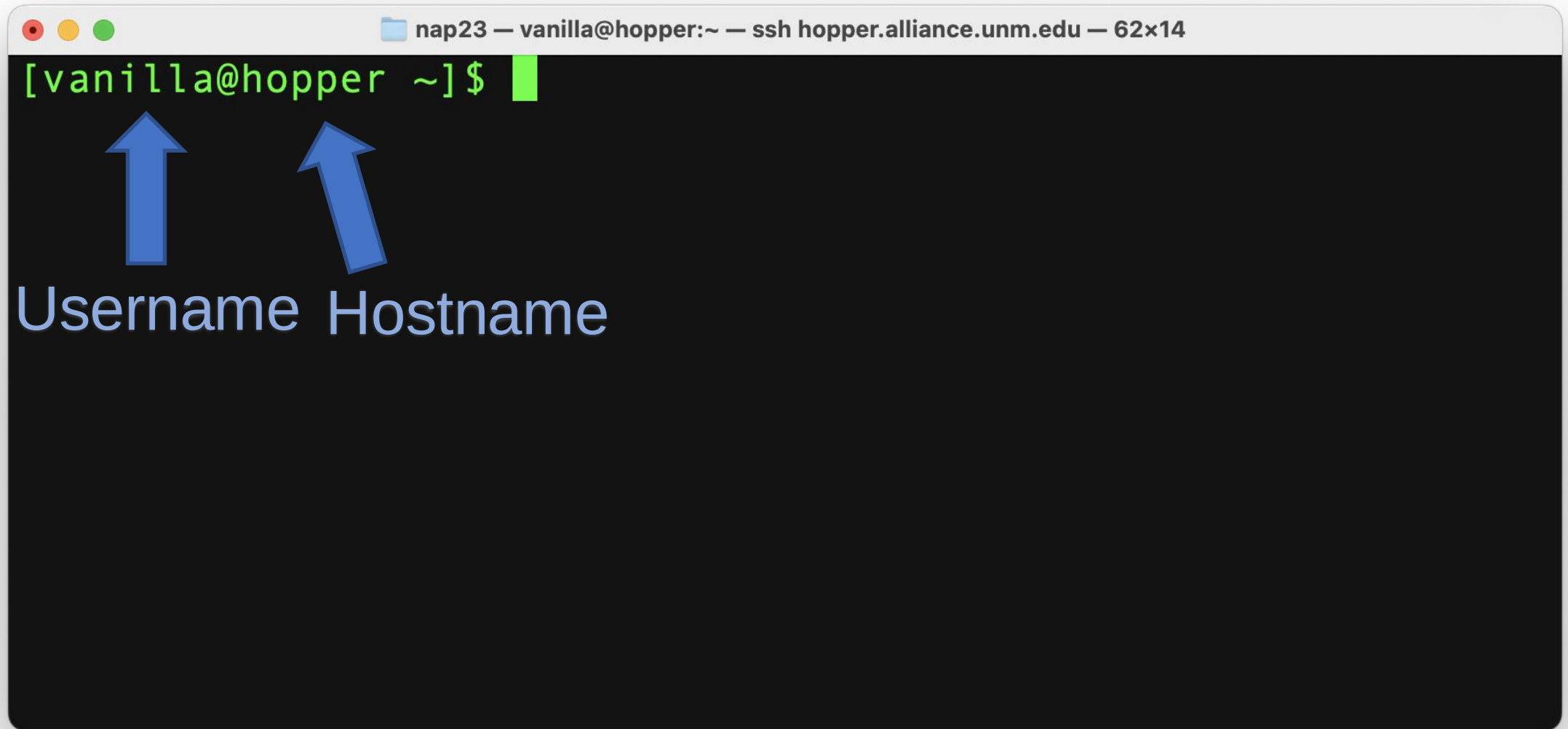


A terminal window titled "nap23 — vanilla@hopper:~ — ssh hopper.alliance.unm.edu — 62x14". The prompt is "[vanilla@hopper ~]\$" in green text on a black background. A blue arrow points from the word "Username" below to the "vanilla" part of the prompt.

```
[vanilla@hopper ~]$
```

Username

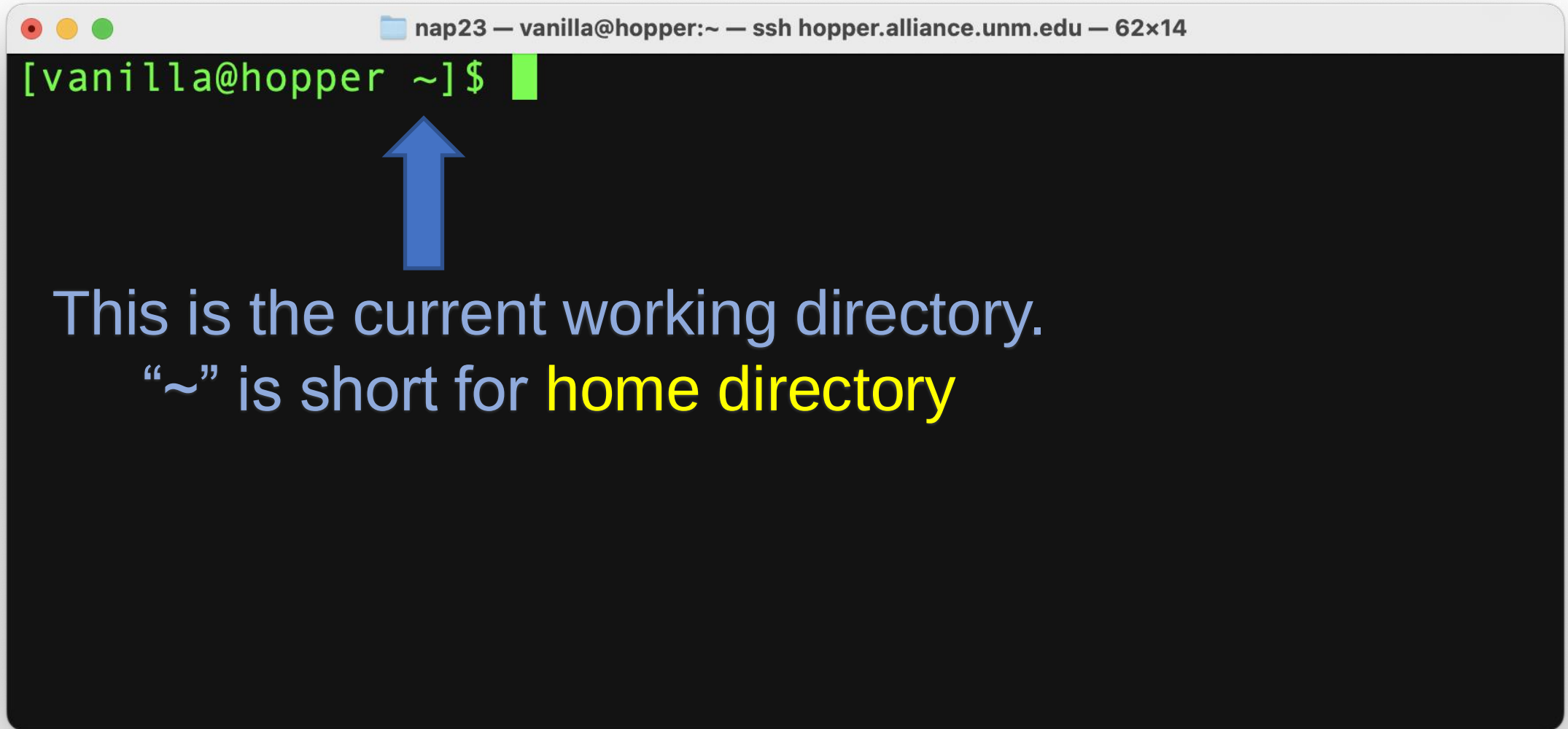
Understanding the BASH prompt...



```
nap23 — vanilla@hopper:~ — ssh hopper.alliance.unm.edu — 62x14
[vanilla@hopper ~]$
```

Username Hostname

Understanding the BASH prompt...



```
nap23 — vanilla@hopper:~ — ssh hopper.alliance.unm.edu — 62x14  
[vanilla@hopper ~]$
```

This is the current working directory.
“~” is short for **home directory**

Understanding the BASH prompt...

```
nap23 — vanilla@hopper:~ — ssh hopper.alliance.unm.edu — 62x14
[vanilla@hopper ~]$
```



“\$” means this user is standard user
(i.e. not a system administrator)

Understanding the BASH prompt...

```
[vanilla@hopper ~]$ passwd
```

Changing password for user vanilla.

Current Password: {text will be hidden}

New password: {text will be hidden}

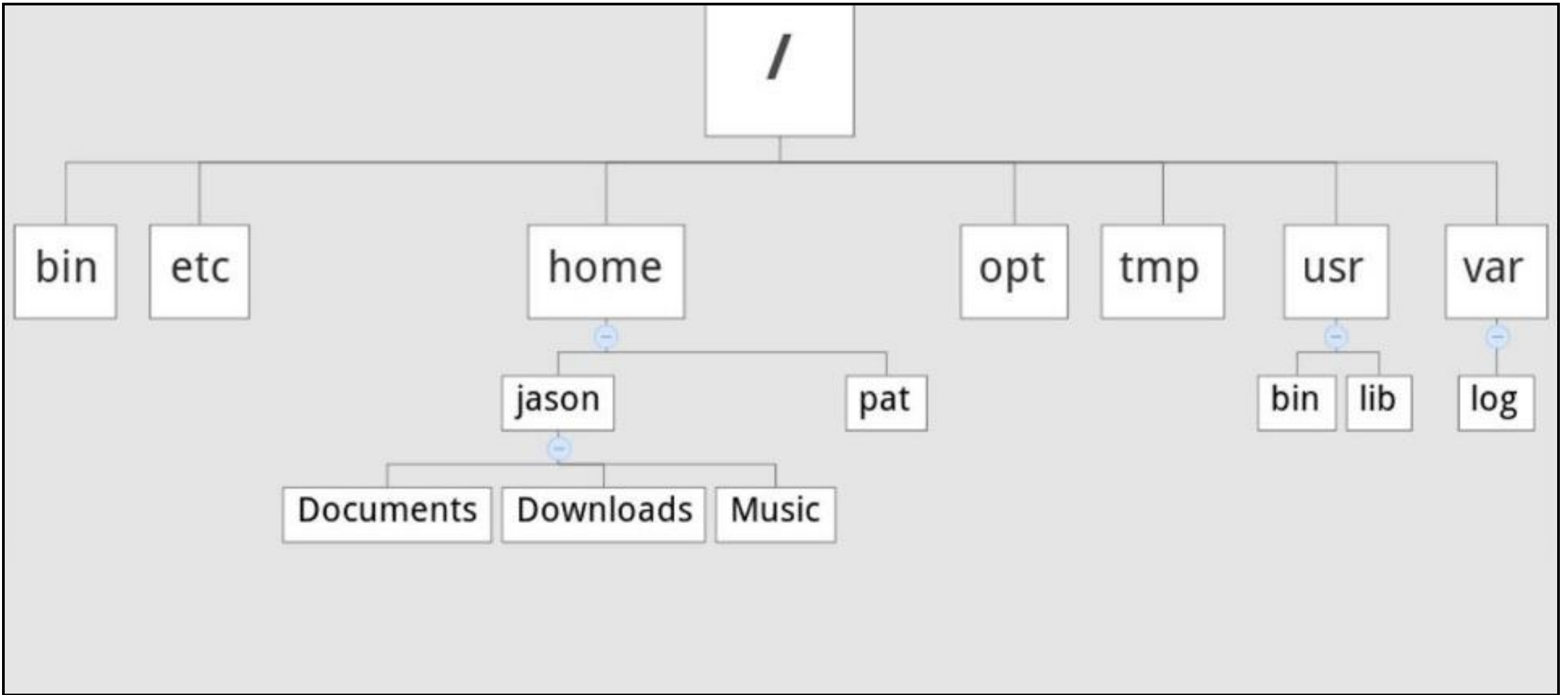
Retype new password: {text will be hidden}

passwd: all authentication tokens updated successfully.

```
[vanilla@hopper ~]$
```



If you used the preset password set your own
password now



Example Filesystem Tree

```
[vanilla@hopper ~]$ pwd  
/users/vanilla  
[vanilla@hopper ~]$
```

Figuring out where you are in the filesystem...

Figuring out where you are in the
filesystem...

Please enter the following command:

```
cp -r /projects/shared/workshops/beginner/mystuff ~/
```

We will come help you if you have any trouble.

(Later I will go over what this command does)



```
[vanilla@hopper ~]$ ls  
mystuff  wheeler-scratch  
[vanilla@hopper ~]$
```

Figuring out where you are in the
filesystem...

```
[vanilla@hopper ~]$ tree
```

```
.  
├── mystuff  
│   ├── myfile1  
│   └── myfile2  
└── wheeler-scratch -> /wheeler/scratch/vanilla
```

```
2 directories, 2 files
```

```
[vanilla@hopper ~]$
```

Figuring out where you are in
the filesystem...

```
[vanilla@hopper ~]$ tree
```

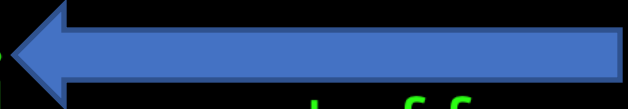
```
.
├── mystuff
│   ├── myfile1
│   └── myfile2
└── wheeler-scratch -> /wheeler/scratch/vanilla
```

```
2 directories, 2 files
```

```
[vanilla@hopper ~]$
```

Figuring out where you are in
the filesystem...

```
[vanilla@hopper ~]$ tree
```

 This . means the current directory

```
.  
├── mystuff  
│   ├── myfile1  
│   └── myfile2  
└── wheeler-scratch -> /wheeler/scratch/vanilla
```

```
2 directories, 2 files
```

```
[vanilla@hopper ~]$
```

Figuring out where you are in
the filesystem...

“Absolute” paths vs “relative” paths

- A path is a list of directories and/or files. It is a path through the directory tree that tells one how to get somewhere in the filesystem.
- An absolute path tells one how to get to the destination from starting from the root of the filesystem. E.g “/users/vanilla/mystuff/”
- A relative path specifies how to get there *starting from the current working directory*. E.g vanilla/mystuff/

```
[vanilla@hopper ~]$ ls mystuff/  
myfile1  myfile2  
[vanilla@hopper ~]$ █
```

Figuring out where you
going...

```
[vanilla@hopper ~]$ ls /users/vanilla/mystuff  
myfile1  myfile2  
[vanilla@hopper ~]$
```

Figuring out where you
going...

```
[vanilla@hopper ~]$ ls ./mystuff/
```

```
myfile1 myfile2
```

```
[vanilla@hopper ~]$ ls ~/mystuff/
```

```
myfile1 myfile2
```

```
[vanilla@hopper ~]$
```

Figuring out where you
going...

```
[vanilla@hopper ~]$ ls -a
.      .modulesbeginenv
..     mystuff
.addressbook  .oracle_jre_usage
.addressbook.lu .pinerc
.bashrc      .pki
.cache       .rhosts
.comsol      .shosts
.config      .spack
.flexlmrc    .ssh
```

Figuring out where you
going...

```
[vanilla@hopper ~]$ ls -l
total 4
drwxr-xr-x 2 vanilla users 4096 Jun 14 22:05 mystuff
lrwxrwxrwx 1 vanilla users 24 Jun 14 21:20 wheeler-scratch -> /wheeler/scratch/vanilla
[vanilla@hopper ~]$
```

Figuring out where you
going...

```
[vanilla@hopper ~]$ ls -l mystuff/  
total 473704  
-rw-r--r-- 1 vanilla users 483165473 Jun 14 23:20 myfile1  
-rw-r--r-- 1 vanilla users      0 Jun 14 22:05 myfile2  
[vanilla@hopper ~]$
```

Figuring out where you
going...

```
[vanilla@hopper ~]$ ls -lh mystuff/  
total 463M  
-rw-r--r-- 1 vanilla users 461M Jun 14 23:20 myfile1  
-rw-r--r-- 1 vanilla users  0 Jun 14 22:05 myfile2  
[vanilla@hopper ~]$
```

Figuring out where you
going...

- Now you know how to view the filesystems using bash
- Let's see how to move around and modify the filesystem.
- To move we used the `cd` (change directory) command.
- In bash to move a file we use the `mv` command.
- To copy a file it is `cp`.
- To copy files from CARC to a personal computer use `scp` or `rsync`.

For more information and graphical tools see the “transferring data” video tutorial in [in this playlist](#).

```
[vanilla@hopper ~]$ cd mystuff  
[vanilla@hopper ~/mystuff]$ mv myfile1 myfile0  
[vanilla@hopper ~/mystuff]$ ls  
myfile0 myfile2 myfile3  
[vanilla@hopper ~/mystuff]
```

Modifying the filesystem...
moving a file.

```
[vanilla@hopper ~/mystuff]$ cp myfile0 myfile1  
[vanilla@hopper ~/mystuff]$
```



Source



Destination

```
[vanilla@hopper ~/mystuff]$ ls  
myfile0 myfile1 myfile2 myfile3  
[vanilla@hopper ~/mystuff]$
```

Modifying the filesystem...
copying a file.

```
[vanilla@hopper ~/mystuff]$ mkdir mynewdir  
[vanilla@hopper ~/mystuff]$
```



New directory name

Modifying the filesystem...
create a new directory.

```
[vanilla@hopper ~]$ cp -r mystuff mystuff2  
[vanilla@hopper ~]$
```



Source



Destination

```
[vanilla@hopper ~]$ ls  
mystuff mystuff2 wheeler-scratch
```

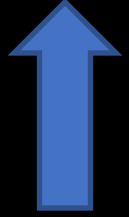
Copying a whole directory
tree...

```
[vanilla@hopper ~]$ exit
```

```
CARCWS-01:~ vanilla$ scp vanilla@hopper.alliance.unm.edu:~/mystuff/myfile3 Desktop/
```



Source



Destination

```
(vanilla@hopper.alliance.unm.edu) Password:  
myfile3          100% 40   2.0KB/s  00:00
```

Copying data to a personal
computer from CARC...

```
CARCWS-01:~ vanilla$ scp -r vanilla@hopper.alliance.unm.edu:~/mystuff Desktop/
```



Source



Destination

```
(vanilla@hopper.alliance.unm.edu) Password:
```

```
myfile1      100% 1024KB  6.5MB/s  00:00  
myfile2      100% 2048KB 382.5KB/s  00:05  
myfile3      100%  40    3.2KB/s  00:00  
myfile0      100% 1024KB  8.8MB/s  00:00
```

Copying data to a personal
computer from CARC...

```
CARCWS-01:~ vanilla$ scp -r Desktop/mystuff vanilla@hopper.alliance.unm.edu:~/
```



Source



Destination

```
(vanilla@hopper.alliance.unm.edu) Password:
```

```
myfile1      100% 1024KB 591.5KB/s  00:01  
myfile0      100% 1024KB  2.0MB/s  00:00  
myfile2      100% 2048KB  2.1MB/s  00:00  
myfile3      100%  40    2.1KB/s  00:00
```

To copy from a personal
computer to CARC...

```
ssh vanilla@hopper.alliance.unm.edu
```

Log back into hopper...

```
[vanilla@hopper ~]$ file mystuff/myfile0  
mystuff/myfile0: data
```

```
[vanilla@hopper ~]$ file mystuff/myfile3  
mystuff/myfile3: ASCII text
```

Figuring out file types ...

```
[vanilla@hopper ~]$ cat mystuff/myfile3
```

Welcome to the CARC Beginner's Workshop

Text files ...

[vanilla@hopper ~]\$ nano mystuff/myfile3

```
Matthew — vanilla@wheeler:~ — ssh wheeler — 71x23
GNU nano 2.3.1      File: mystuff/myfile3

elcome to the CARC Beginner's Workshop

[ Read 1 line ]
^G Get Help ^O WriteOut ^R Read Fil ^Y Prev Pag ^K Cut Text ^C Cur Pos
^X Exit     ^J Justify  ^W Where Is ^V Next Pag ^U UnCut Te ^T To Spell
```

```
[vanilla@hopper ~]$ date  
Wed Jun 15 03:08:15 MDT 2022
```

```
[vanilla@hopper ~]$ echo Hello from $HOSTNAME  
Hello from hopper
```

```
[vanilla@hopper ~]$ hostname  
hopper
```

Programs we will use as
examples...

```
[vanilla@hopper ~]$ nano myscript.bash
```

BASH Script

File Edit Options Buffers Tools Sh-Script Help

```
#!/bin/bash
```

```
echo Hello from $HOSTNAME
```

```
date
```

Shell Scripts

```
[vanilla@hopper ~]$ nano myscript.bash
```

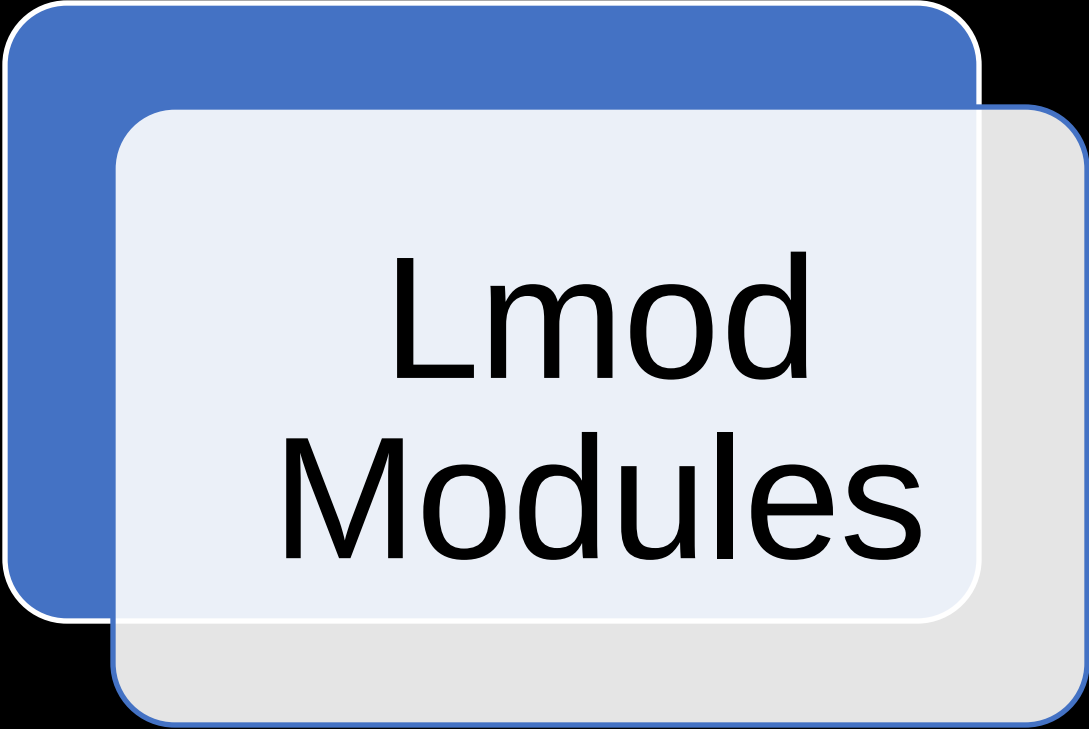
```
[vanilla@hopper ~]$ bash myscript.bash
```

```
Hello from hopper
```

```
Wed Mar 26 16:17:52 MDT 2025
```

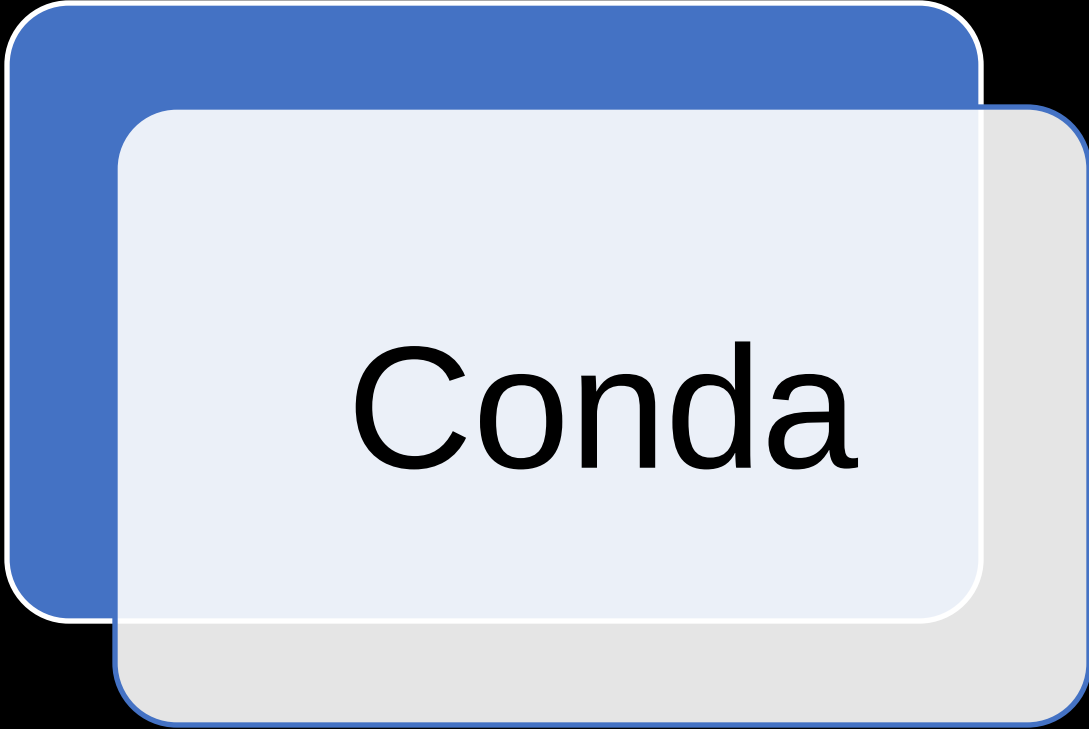
BASH Script

Software Access

A graphic consisting of three overlapping rounded rectangles. The top rectangle is blue, the middle one is light blue, and the bottom one is light gray. The text "Lmod Modules" is centered in the light blue rectangle.

**Lmod
Modules**

Installed Systemwide by CARC personnel
Email help@carc.unm.edu

A graphic consisting of three overlapping rounded rectangles. The top rectangle is blue, the middle one is light blue, and the bottom one is light gray. The text "Conda" is centered in the light blue rectangle.

Conda

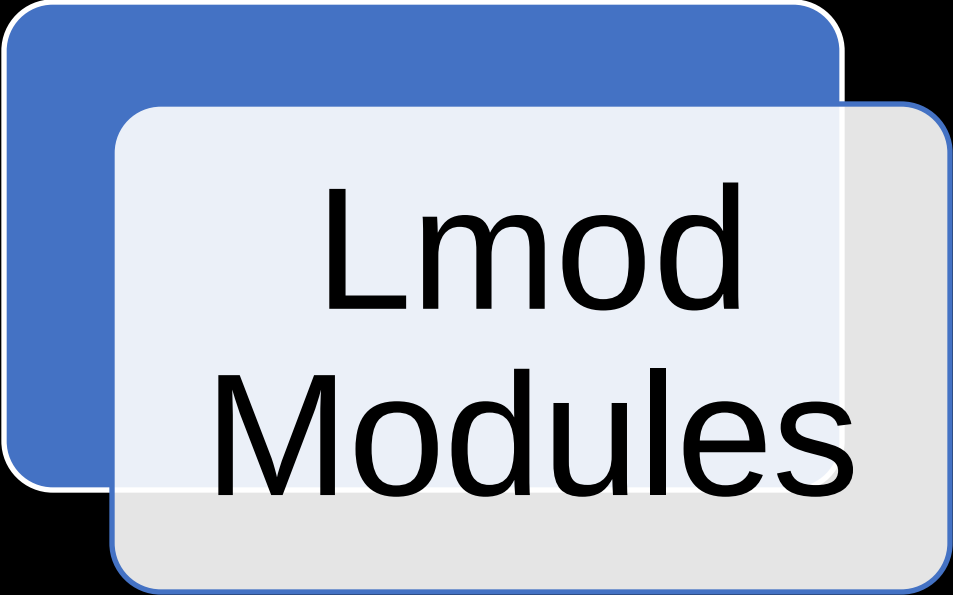
Installed in your home dir yourself

```
[vanilla@hopper ~]$ module spider matlab
```

```
matlab:
```

Versions:

- matlab/R2017a
- matlab/R2018b
- matlab/R2019a
- matlab/R2020a
- matlab/R2021a
- matlab/R2021b
- matlab/R2022a
- matlab/R2023a
- matlab/R2023b

A graphic consisting of two overlapping rounded rectangles. The front rectangle is light blue with a dark blue border and contains the text 'Lmod Modules' in a large, black, sans-serif font. The back rectangle is a darker blue and is partially obscured by the front one.

Lmod Modules

Getting access to software...

```
[vanilla@hopper ~]$ module load matlab/R2021a
```

```
Lmod has detected the following error: Matlab may only be run on compute  
nodes. hopper is not a compute node. Exiting...
```

```
While processing the following module(s):
```

```
Module fullname Module Filename
```

```
-----
```

```
matlab/R2021a /opt/local/modules/matlab/R2021a.lua
```

Getting access to software...

```
[vanilla@hopper ~]$ module load matlab/R2021a
```

```
Lmod has detected the following error: Matlab may only be run on compute  
nodes. hopper is not a compute node. Exiting...
```

```
While processing the following module(s):
```

```
Module fullname  Module Filename
```

```
-----
```

```
matlab/R2021a    /opt/local/modules/matlab/R2021a.lua
```

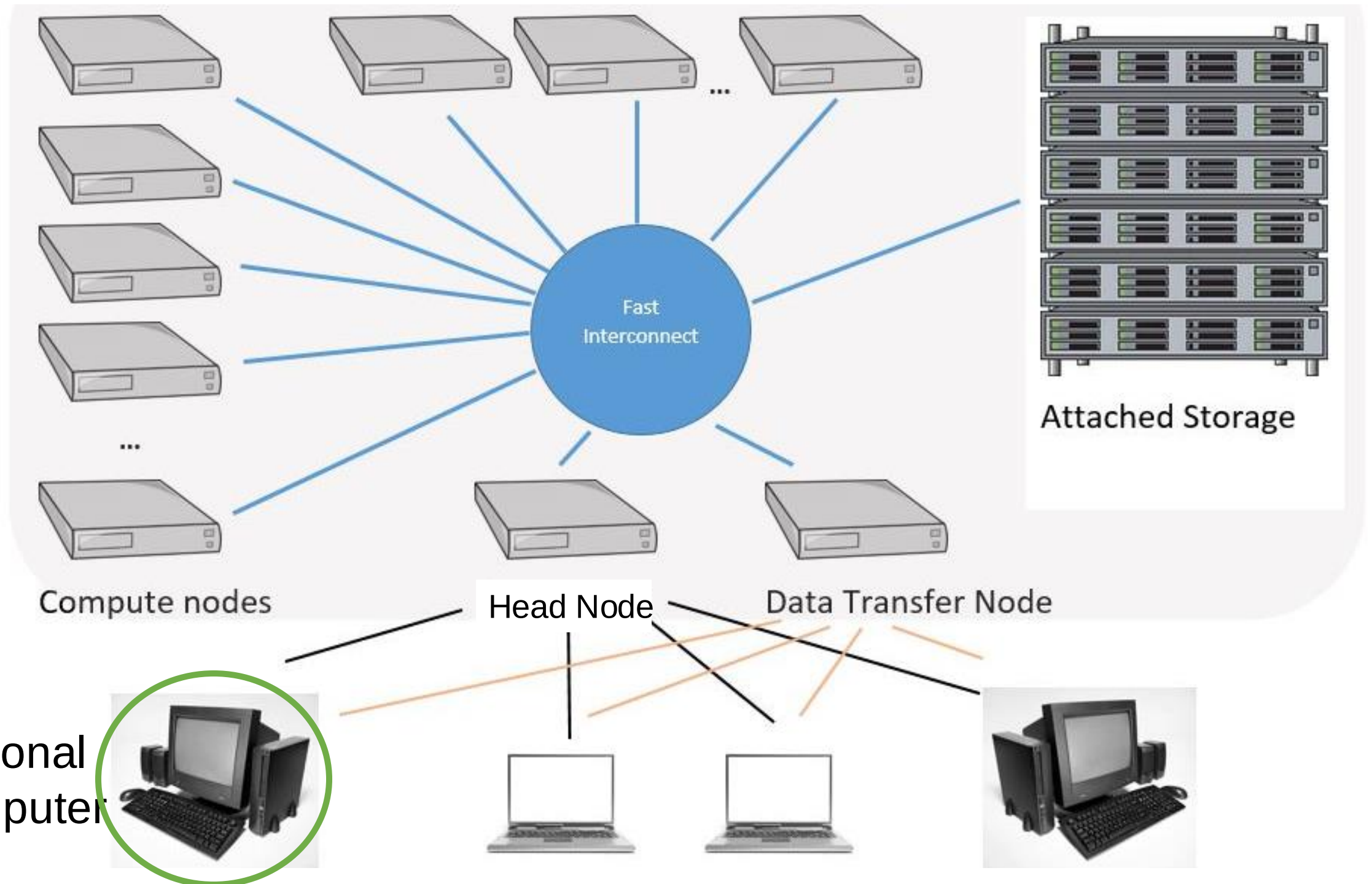
What is a compute node?

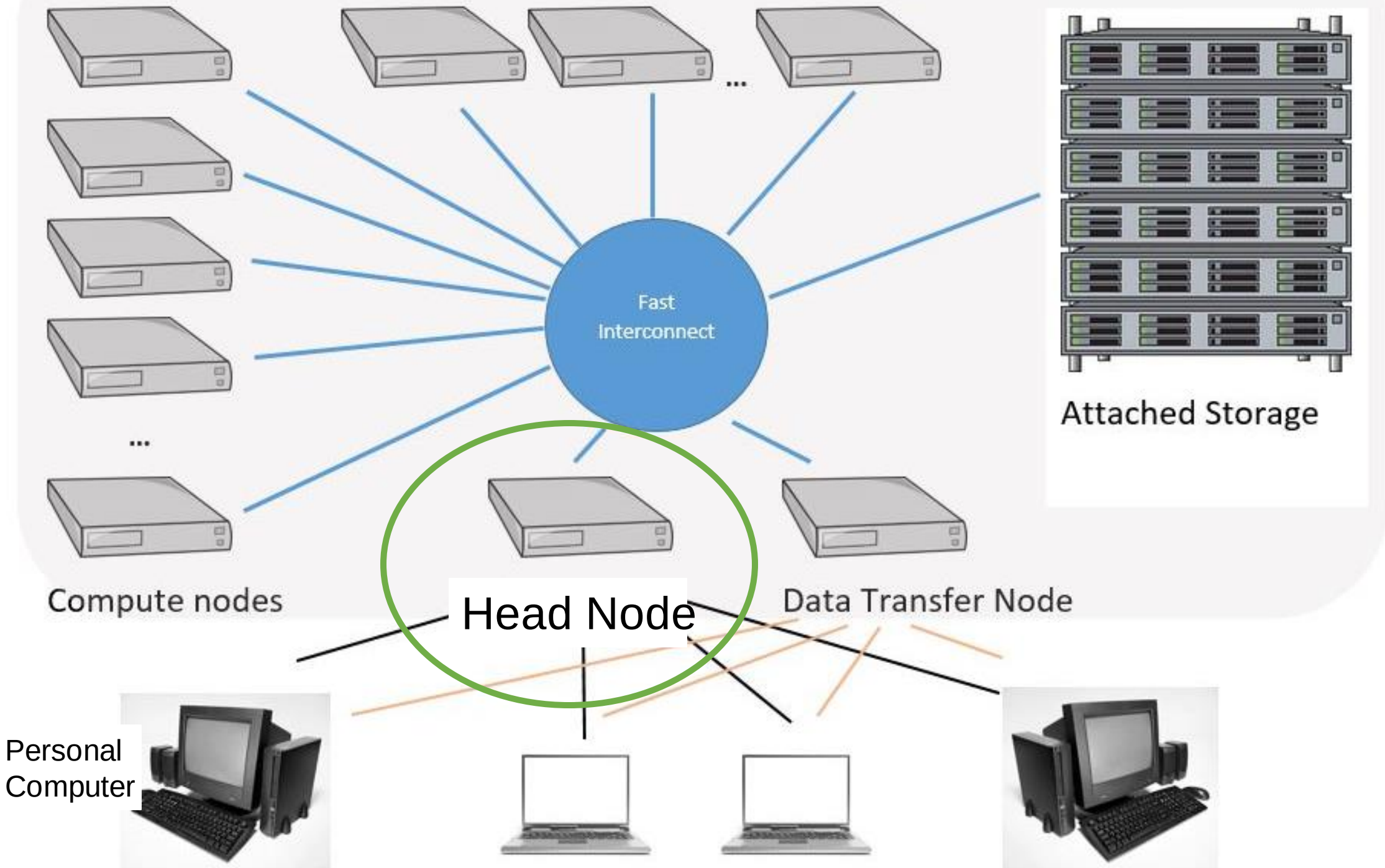
Getting access to software...

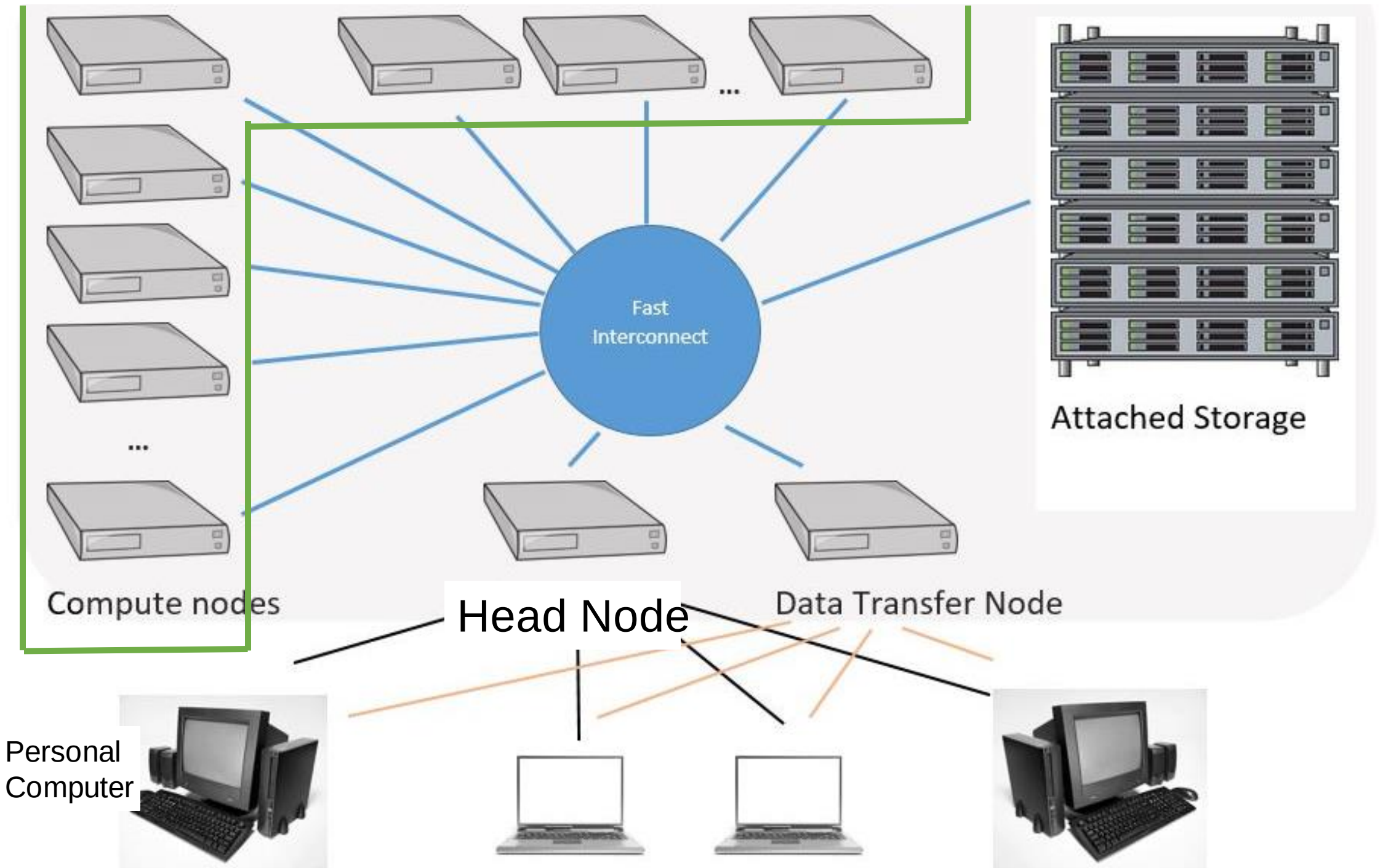
A photograph of a server room with rows of server racks. The racks are filled with electronic components and have many small, colorful indicator lights. The room is dimly lit with a blue hue. A circular dark overlay is positioned on the left side of the image, containing the text 'HPC Cluster' in white.

HPC Cluster

Personal
Computer







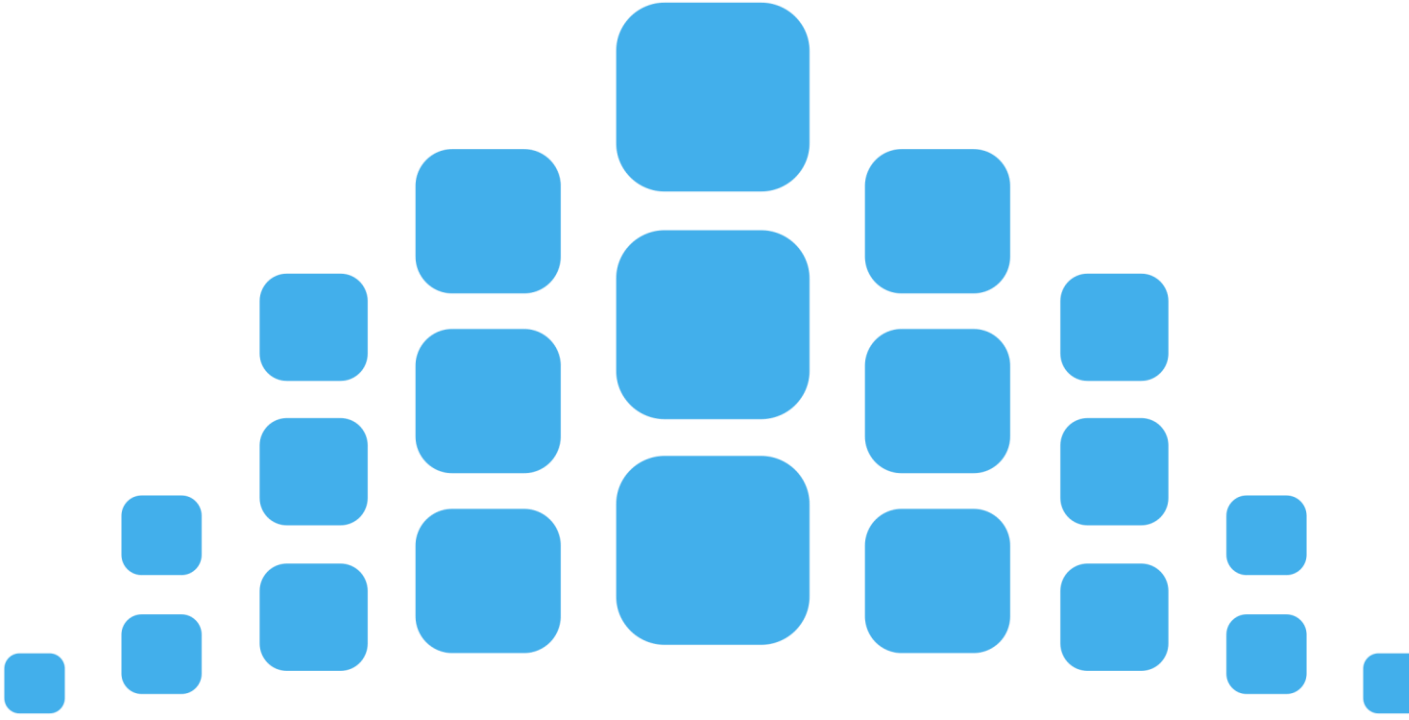


Never run computations on the head node

(This can make it so no one else can login)

Always use compute nodes

Simple Linux
Utility for
Resource
Management



slurm
workload manager

S



Simple Linux
or
ce
ment

workload manager

[vanilla@hopper ~]\$ qgrok

partition	nodes					total	total	free	total	free	CPU	RAM/node	time	CPU	GPU	RAM
name	jobs	free	busy	down	nodes	CPUs	CPUs	CPUs	GPUs	GPUs	/node		limit	limit	limit	limit
general	4	4	6	0	10	320	156	0	0	32	93G	2d	128	0	372G	
debug	1	1	1	0	2	64	63	0	0	32	93G	4h	8	0	25G	
totals:	5	5	7	0	12	384	219	0	0							

[vanilla@hopper ~]\$ qgrok

partition	nodes				total	total	free	total	free	CPU	RAM/node	time	CPU	GPU	RAM
name	jobs	free	busy	down	nodes	CPU	CPU	GPU	GPU	GPUs	/node	limit	limit	limit	limit
general	4	4	6	0	10	320	156	0	0	32	93G	2d	128	0	372G
debug	1	1	1	0	2	64	63	0	0	32	93G	4h	8	0	25G
condo	29	30	25	0	55	1760	1060	37	8	32	94G-1.5T	2d	512	4	1.5T
bugs	0	0	2	0	2	64	0	0	0	32	93G	1w	64	0	
pcnc	0	0	2	0	2	64	0	0	0	32	93G	1w	64	0	
pathogen	0	1	0	0	1	32	32	0	0	32	93G	1w	32	0	
tc	8	1	9	0	10	320	104	0	0	32	93G-1.5T	1w	320	0	
gold	0	2	0	0	2	64	64	0	0	32	93G	1w	64	0	
fishgen	1	0	1	0	1	32	12	0	0	32	377G	1w	32	0	
neuro-hsc	0	14	0	0	14	448	448	0	0	32	93G	1w	448	0	
pna	0	1	0	0	1	32	32	0	0	32	93G	1w	32	0	
geodef	0	4	0	0	4	128	128	0	0	32	504G	1w	128	0	
cup-ecs	0	0	2	0	2	64	0	13	8	32	188G	1w	64	13	
tid	0	0	1	0	1	32	0	2	0	32	188G	1w	32	2	
biocomp	0	1	0	0	1	32	32	1	0	32	188G	1w	32	1	
chakra	0	0	1	0	1	32	0	1	0	32	188G	1w	32	1	
quark	0	3	7	0	10	320	112	20	0	32	377G	1w3d	320	20	
toadpole	0	1	0	0	1	32	32	0	0	32	94G	1w	32	0	
insar	0	2	0	0	2	64	64	0	0	32	504G	1w	64	0	
totals:	34	35	32	0	67	2144	1279	37	8						

[vanilla@hopper ~]\$ qgrok

partition	nodes	nodes	nodes	nodes	total	total	free	total	free	CPU	RAM/node	time	CPU	GPU	RAM
name	jobs	free	busy	down	nodes	CPU	CPU	GPU	GPU	/node	limit	limit	limit	limit	
general	4	4	6	0	10	320	156	0	0	32	93G	2d	128	0	372G
debug	1	1	1	0	2	64	63	0	0	32	93G	4h	8	0	25G
condo	29	30	25	0	55	1760	1060	37	8	32	94G-1.5T	2d	512	4	1.5T
bugs	0	0	2	0	2	64	0	0	0	32	93G	1w	64	0	
pcnc	0	0	2	0	2	64	0	0	0	32	93G	1w	64	0	
pathogen	0	1	0	0	1	32	32	0	0	32	93G	1w	32	0	
tc	8	1	9	0	10	320	104	0	0	32	93G-1.5T	1w	320	0	
gold	0	2	0	0	2	64	64	0	0	32	93G				
fishgen	1	0	1	0	1	32	12	0	0	32	377				
neuro-hsc	0	14	0	0	14	448	448	0	0	32					
pna	0	1	0	0	1	32	32	0	0	32	93				
geodef	0	4	0	0	4	128	128	0	0	32					
cup-ecs	0	0	2	0	2	64	0	13	8	32	1				
tid	0	0	1	0	1	32	0	2	0	32	1880				
biocomp	0	1	0	0	1	32	32	1	0	32					
chakra	0	0	1	0	1	32	0	1	0	32	18				
quark	0	3	7	0	10	320	112	20	0	32					
toadpole	0	1	0	0	1	32	32	0	0	32					
insar	0	2	0	0	2	64	64	0	0	32	50				
totals:	34	35	32	0	67	2144	1279	37	8						

Open partitions
everyone with a

Purchased by the

Open partitions for use by everyone with a CARC account.

Purchased by the Office for the Vice President for Research.

```
[vanilla@hopper ~]$ sinfo --partition debug
```

```
PARTITION AVAIL TIMELIMIT NODES STATE NODELIST
```

```
debug      up    4:00:00    1  mix hopper011
```

```
debug      up    4:00:00    1  idle hopper012
```

sinfo reports information about
partitions

```
[vanilla@hopper ~]$ sinfo --partition debug
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
debug      up    4:00:00    1  mix hopper011
debug      up    4:00:00    1  idle hopper012
```

The debug queues are intended
for testing your programs.

And for interactive jobs.

```
[vanilla@hopper ~]$ sinfo --partition debug
```

```
PARTITION AVAIL TIMELIMIT NODES STATE NODELIST
```

```
debug      up    4:00:00    1  mix hopper011
```

```
debug      up    4:00:00    1  idle hopper012
```



Name

```
[vanilla@hopper ~]$ sinfo --partition debug
```

```
PARTITION AVAIL TIMELIMIT NODES STATE NODELIST
```

```
debug      up    4:00:00    1  mix hopper011
```

```
debug      up    4:00:00    1  idle hopper012
```



You can run a “job” for up to 4 hrs.

```
[vanilla@hopper ~]$ sinfo --partition debug
```

```
PARTITION AVAIL TIMELIMIT NODES STATE NODELIST
```

```
debug      up    4:00:00    1  mix hopper011
```

```
debug      up    4:00:00    1  idle hopper012
```



There are two nodes in this partition.

```
[vanilla@hopper ~]$ sinfo --partition debug
```

PARTITION	AVAIL	TIMELIMIT	NODES	STATE	NODELIST
-----------	-------	-----------	-------	-------	----------

debug	up	4:00:00	1	mix	hopper011
-------	----	---------	---	-----	-----------

debug	up	4:00:00	1	idle	hopper012
-------	----	---------	---	------	-----------



The state of the nodes in the partition

```
[vanilla@hopper ~]$ sinfo --partition debug
```

```
PARTITION AVAIL TIMELIMIT NODES STATE NODELIST
```

```
debug      up    4:00:00    1  mix hopper011
```

```
debug      up    4:00:00    1  idle hopper012
```



The name of the nodes in the
partition

```
[vanilla@hopper ~]$ sinfo --partition general
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
general*   up 2-00:00:00    5 alloc hopper[001-009]
general*   up 2-00:00:00    4 idle  hopper010
```



Hopper001 through 009 are running jobs.

Hopper010 is waiting to be used.

```
[vanilla@hopper ~]$ hostname  
hopper  
[vanilla@hopper ~]$
```



Running on the Head Node.

The head node's name is "hopper".

```
[vanilla@hopper ~]$ srun --partition debug hostname
```



Tell slurm to run a program
on a compute node...

```
[vanilla@hopper ~]$ srun --partition debug hostname
```



Run the program on a
compute node in the
debug partition.

```
[vanilla@hopper ~]$ srun --partition debug hostname
```



The program
to run.

```
[vanilla@hopper ~]$ srun --partition debug hostname
```

```
srun: Account not specified in script or ~/.default_slurm_account, using latest project
```

```
You have not been allocated GPUs. To request GPUs, use the -G option in your submission script.
```

```
hopper011
```

```
[vanilla@hopper ~]$ queue
```

```
[vanilla@hopper ~]$ squeue
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	ODELIST(REASON)
4314	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4315	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4317	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4318	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4319	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4320	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4321	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4322	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4323	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4324	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4325	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4326	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4328	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4329	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4330	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4331	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4332	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4333	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4334	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4335	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4336	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	
4337	general	PRE erowland	PD	0:00	2	(QOSMaxCpuPerUserLimit)	

```
[vanilla@hopper ~]$ squeue
```

JOBID	PARTITION	NAME	USER	ST	TIME
4314	general	PRE	erowland	PD	0:00
4315	general	PRE	erowland	PD	0:00
4317	general	PRE	erowland	PD	0:00
4318	general	PRE	erowland	PD	0:00
4319	general	PRE	erowland	PD	0:00
4320	general	PRE	erowland	PD	0:00
4321	general	PRE	erowland	PD	0:00
4322	general	PRE	erowland	PD	0:00
4323	general	PRE	erowland	PD	0:00
4324	general	PRE	erowland	PD	0:00
4325	general	PRE	erowland	PD	0:00
4326	general	PRE	erowland	PD	0:00
4328	general	PRE	erowland	PD	0:00
4329	general	PRE	erowland	PD	0:00
4330	general	PRE	erowland	PD	0:00
4331	general	PRE	erowland	PD	0:00
4332	general	PRE	erowland	PD	0:00
4333	general	PRE	erowland	PD	0:00
4334	general	PRE	erowland	PD	0:00
4335	general	PRE	erowland	PD	0:00
4336	general	PRE	erowland	PD	0:00
4337	general	PRE	erowland	PD	0:00

PD means programs
that are waiting their
turn.

Shows you what the slurm
scheduler is doing right now.

Here we can see that user
'erowland' has a lot of programs
waiting to run.

```
[vanilla@hopper ~]$ squeue -t R --all
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
4405	condo	2ndMA	mfricke	R	1-07:48:30	6	hopper[031-036]
5208	condo	NN	kgu	R	5:48:49	1	hopper015
5210	condo	NN	kgu	R	6:30:13	1	hopper014
5209	condo	NN	kgu	R	6:31:13	1	hopper013
5206	condo	NN	kgu	R	6:32:13	1	hopper051
5207	condo	NN	kgu	R	6:32:13	1	hopper052
5205	condo	NN	kgu	R	6:32:43	1	hopper028
4595	cup-ecs	golConfi	aalasand	R	2-06:51:59	1	hopper050
4594	cup-ecs	golConfi	aalasand	R	2-06:52:03	1	hopper049
5120	general	jupyterh	jacobm	R	11:45:47	1	hopper007
4313	general	PRE	erowland	R	1:17:29	2	hopper[003-004]
5111	general	1stMA	mfricke	R	11:15:28	2	hopper[005-006]
5025	general	c2n	jxzuo	R	1:50	1	hopper001
5024	general	c2n	jxzuo	R	31:28	1	hopper002
5203	general	NN	kgu	R	6:37:50	1	hopper009
5201	general	NN	kgu	R	6:38:14	1	hopper008
4390	tc	UCsTpCyd	lepluart	R	2-15:18:18	3	hopper[018-020]
5198	tc	NN	kgu	R	6:40:19	1	hopper030
5196	tc	NN	kgu	R	6:40:31	1	hopper029

```
[vanilla@hopper ~]$ squeue -t R --all
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
4405	condo	2ndMA	mfricke	R	1-07:48:30	6	hopper[031-036]
5208	condo	NN	kgu	R	5:48:49	1	hopper015
5210	condo	NN	kgu	R	6:30:13	1	hopper014
5209	condo	NN	kgu	R	6:31:13	1	hopper013
5206	condo	NN	kgu	R	6:32:13	1	hopper051
5207	condo	NN	kgu	R	6:32:13	1	hopper052
5205	condo	NN	kgu	R	6:32:43	1	hopper028
4595	cup-ecs	golConfi	aalasand	R	2-06:51:59	1	hopper050
4594	cup-ecs	golConfi	aalasand	R	2-06:52:03	1	hopper049
5120	general	jupyterh	jacobm	R	11:45:47	1	hopper007
4313	general	PRE	erowland	R	1:17:29	2	hopper[003-C
5111	general	1stMA	mfricke	R	11:15:28	2	hopper[005-006]
5025	general	c2n	jxzuo	R	1:50	1	hopper001
5024	general	c2n	jxzuo	R	31:28	1	hopper002
5203	general	NN	kgu	R	6:37:50	1	hopper009
5201	general	NN	kgu	R	6:38:14	1	hopper008
4390	tc	UCsTpCyd	lepluart	R	2-15:18:18	3	hopper[018-020]
5198	tc	NN	kgu	R	6:40:19	1	hopper030
5196	tc	NN	kgu	R	6:40:31	1	hopper029



**Only one job
running**

```
[vanilla@hopper ~]$ srun --partition debug --ntasks 8 hostname
```

```
srun: Account not specified in script or ~/.default_slurm_account, using latest project
```

```
hopper011
```

```
hopper011
```

```
hopper011
```

```
You have not been allocated GPUs. To request GPUs, use the -G option in your submission script.
```

```
hopper011
```

```
hopper011
```

```
hopper011
```

```
hopper011
```

```
hopper011
```

You ran eight **copies** of your program.

ntasks is the number of copies to run.

```
[vanilla@hopper ~]$ srun --partition debug --ntasks 8 hostname
```

```
srun: Account not specified in script or ~/.default_slurm_account, using latest project
```

```
hopper011
```

```
hopper011
```

```
hopper011
```

```
You have not been allocated GPUs. To request GPUs, use the -G option in your submission script.
```

```
hopper011
```

```
hopper011
```

```
hopper011
```

```
hopper011
```

```
hopper011
```

By default, each task (copy of your program) is allowed to use one CPU.

Many programs are able to use more than one CPU at a time.

```
[vanilla@hopper ~]$ srun --partition debug --ntasks 2 --cpus-per-task 2 hostname  
srun: Account not specified in script or ~/.default_slurm_account, using latest project  
You have not been allocated GPUs. To request GPUs, use the -G option in your submission script.  
hopper011  
hopper011
```

**Here we are telling SLURM to
run 2 copies of our program
and let each copy of our
program use 2 CPUs.**

```
[vanilla@hopper ~]$ srun --partition debug --mem 4G
```

```
--ntasks 2 --cpus-per-task 2 hostname
```

```
srun: Account not specified in script or ~/.default_slurm_account, using latest project
```

```
hopper012
```

```
hopper012
```

```
You have not been allocated any resources in this submission script.
```

And we can specify how much memory we want.

--mem 4G means give me 4 gigabytes of memory per node.

```
[vanilla@hopper ~]$ srun --partition debug --mem 4G
```

```
--ntasks 2 --cpus-per-task 2 hostname
```

```
srun: Account not specified in script or ~/.default_slurm_account, using latest project
```

```
hopper012
```

```
hopper012
```

```
You have not been allocated any resources.  
submission script.
```

Why does all this matter?

The purpose of SLURM is to provide you the hardware your programs need.

So you have to understand what those requirements are really well.

```
[vanilla@hopper ~]$ srun --partition debug --mem 4G
```

```
--ntasks 2 --cpus-per-task 2 hostname
```

```
srun: Account not specified in script or ~/.default_slurm_account, using latest project
```

```
hopper012
```

```
hopper012
```

```
You have not been allocated any resources in this  
submission script.
```

- 1) Can my program use multiple CPUs?
- 2) How much memory does my program need?
- 3) Can my program use multiple compute nodes (MPI*, GNU Parallel*)?
- 4) Can my program use GPUs?

Interactive vs Batch Mode

Interactive Mode

- Everything so far has been interactive. You request hardware, run your program, and get the output on your screen right away.

Batch Mode

- Most programs at an HPC center are run in “batch” mode.
- Batch mode means we write a shell script that the SLURM scheduler runs for us. The script requests hardware just like we did with salloc and then runs the commands in the script.
- Whatever would have been written to the screen is saved to a file instead.

```
[vanilla@hopper ~]$ git clone https://lobogit.unm.edu/CARC/workshops.git
```

```
Cloning into 'workshops'...
```

```
remote: Enumerating objects: 132, done.
```

```
remote: Counting objects: 100% (75/75), done.
```

```
remote: Compressing objects: 100% (43/43), done.
```

```
remote: Total 132 (delta 33), reused 74 (delta 32), pack-reused 57
```

```
Receiving objects: 100% (132/132), 57.58 KiB | 3.60 MiB/s, done.
```

```
Resolving deltas: 100% (51/51), done.
```

Rather than make you write shell scripts lets just download some we wrote for this workshop...

```
[vanilla@hopper ~]$ tree workshops
```

```
workshops
├── crystallography
│   ├── data
│   │   └── thaumatin.pdb
│   └── slurm
│       ├── integrate_xray_images.slurm
│       └── solve_structure.slurm
├── escape
│   ├── fit_insar_data.ipynb
│   └── kml_functions.py
├── gaussian
│   ├── AcetylChloride.com
│   └── gaussian16_linda_acetylchloride.sh
├── intro_workshop
│   └── code
│       ├── calc_pi_mpi.f90
│       ├── calcPiMPI.py
│       ├── calc_pi_serial.f90
│       └── calcPiSerial.py
...
```

Run tree to see how the workshops directories are organized...

6 directories, 30 files

```
[vanilla@hopper intro_workshop]$ pwd  
/users/vanilla/workshops/intro_workshop  
[vanilla@hopper intro_workshop]$ cat slurm/workshop_example1.sh  
#!/bin/bash  
#SBATCH --partition debug  
#SBATCH --ntasks 4  
#SBATCH --time 00:05:00  
#SBATCH --job-name ws_example  
#SBATCH --mail-user your_username@unm.edu  
#SBATCH --mail-type ALL  
  
srun hostname
```

Let's take a look at the **workshop_example.sh** script in the slurm directory...

```
[vanilla@hopper intro_workshop]$ sbatch slurm/workshop_example1.sh
sbatch: Account not specified in script or ~/.default_slurm_account, using latest project
Submitted batch job 5252
[vanilla@hopper intro_workshop]$
```

We **submit** our slurm
shell script with the
sbatch command.

Workflow

Head Node

User 1

Program A

Script A

User 2

Program B

Script B

Compute Node 01

Compute Node 02

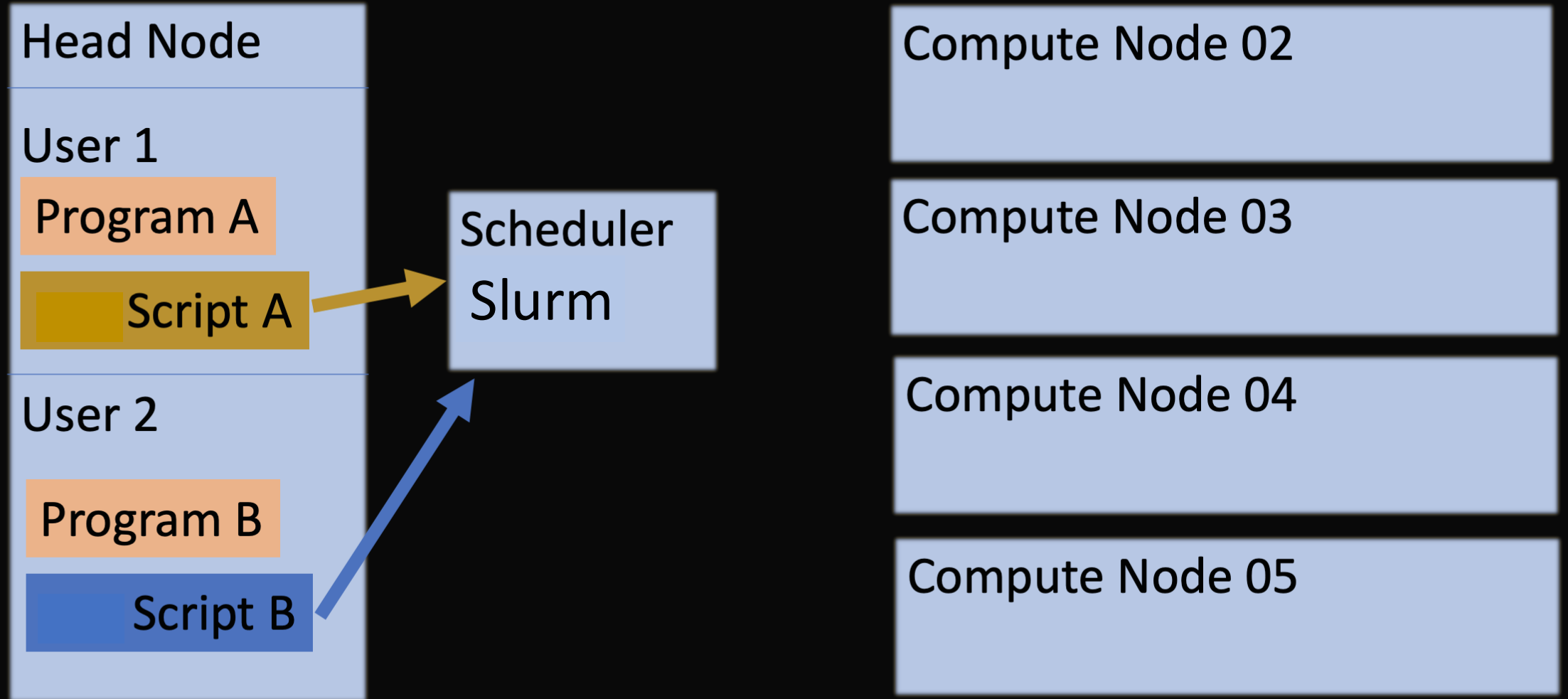
Compute Node 03

Compute Node 04

Compute Node 05

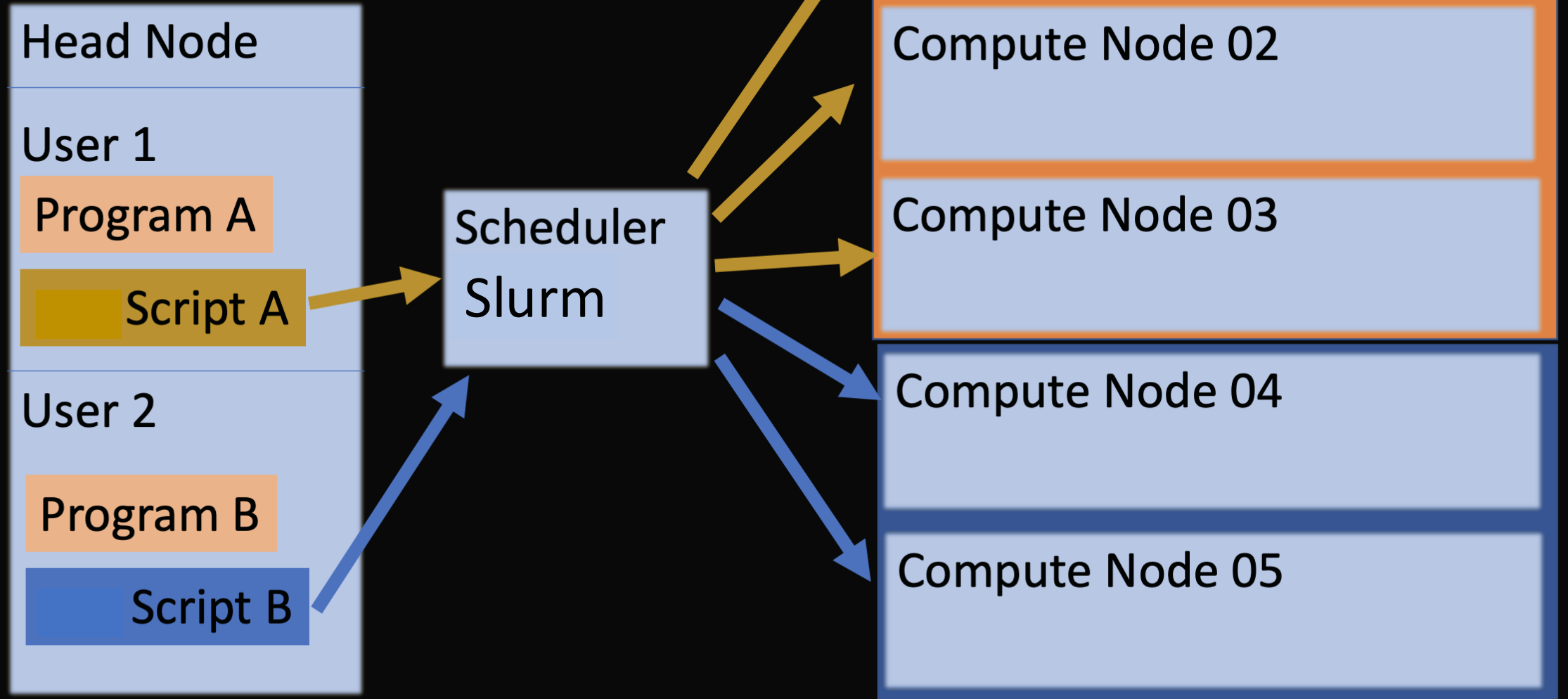
Shared filesystems – All nodes can access the same programs and write output

Workflow



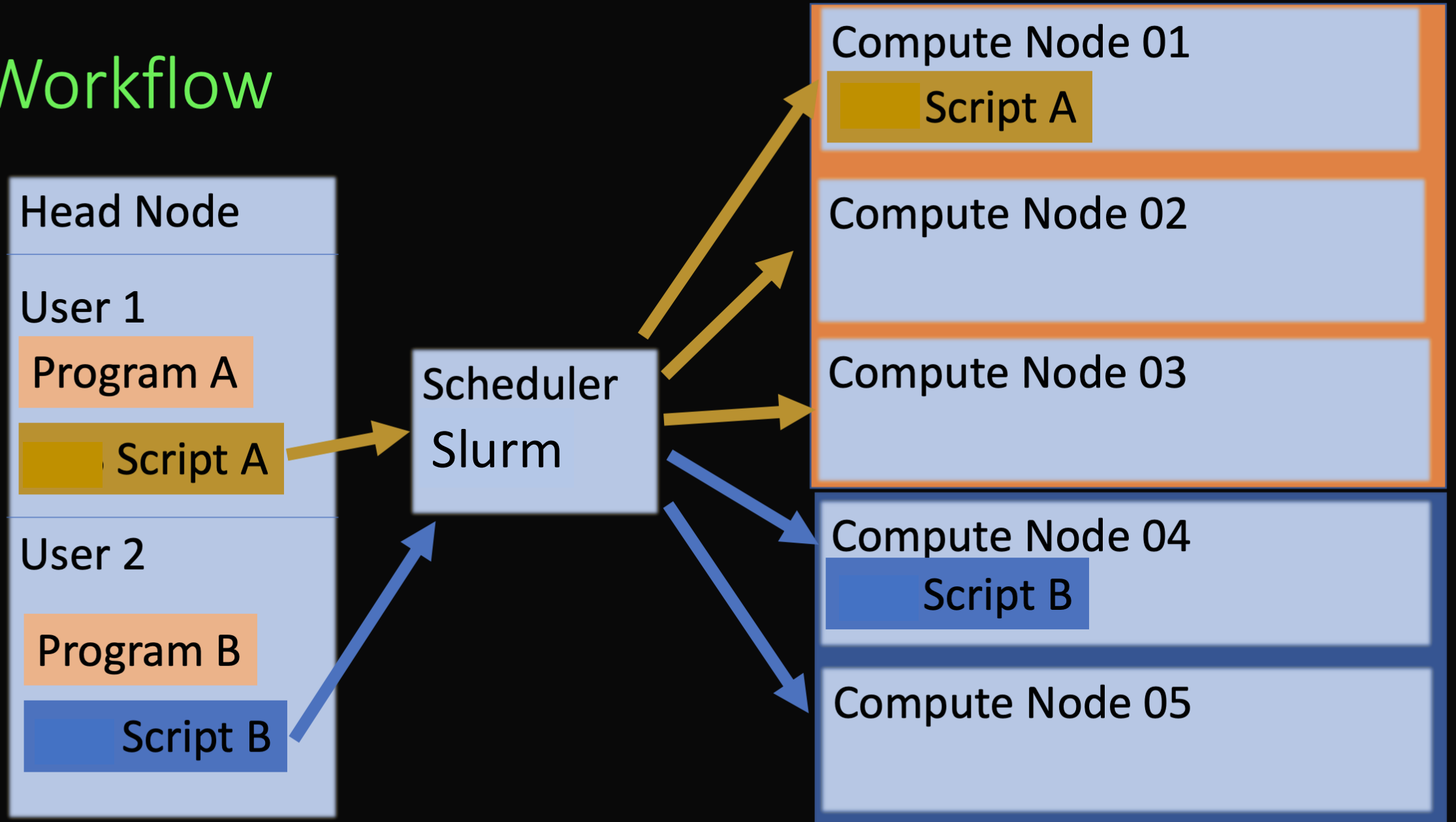
Shared filesystems – All nodes can access the same programs and write output

Workflow



Shared filesystems – All nodes can access the same programs and write output

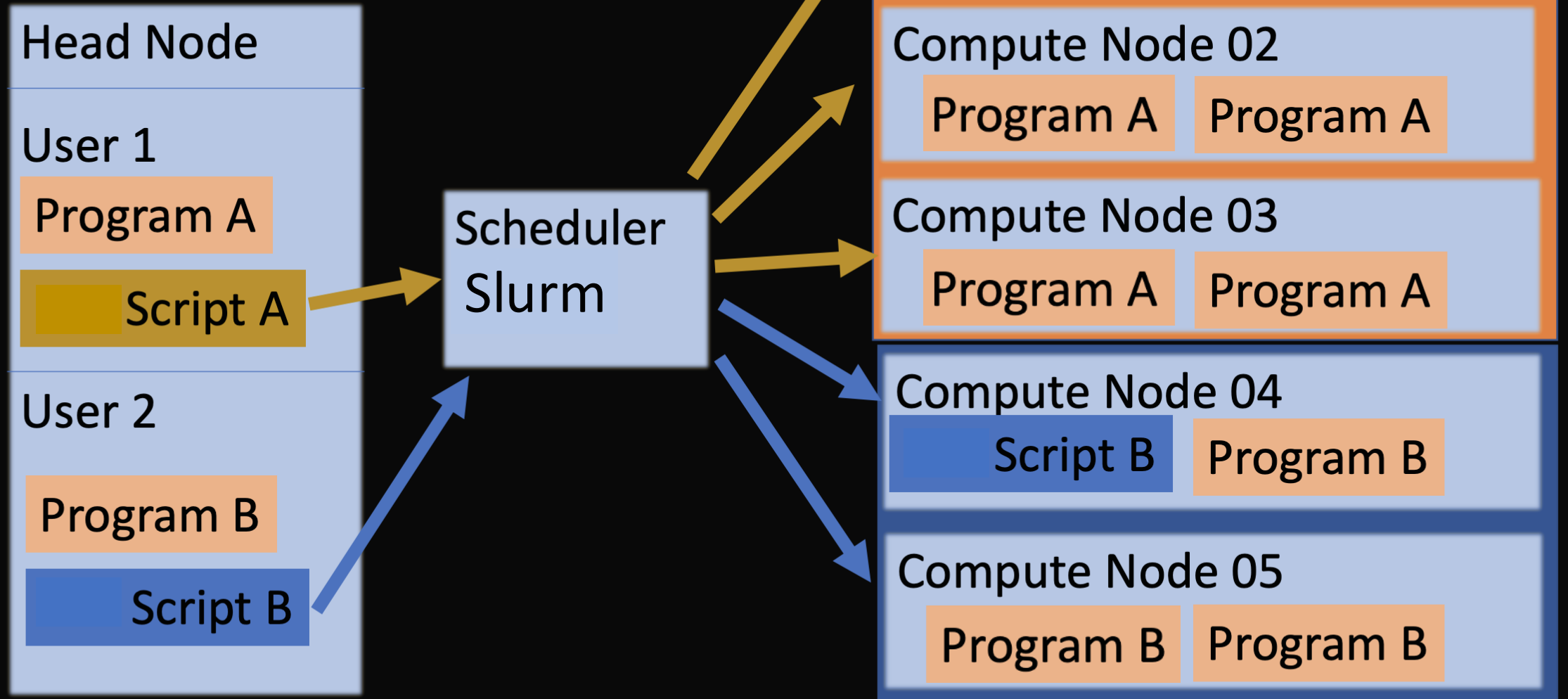
Workflow



Shared filesystems – All nodes can access the same programs and write output

Workflow

We need something in the script to run the program on all the nodes. E.g. `srun`.



Shared filesystems – All nodes can access the same programs and write output

```
[vanilla@hopper intro_workshop]$ ls  
code data pbs slurm slurm-5252.out
```

The **hostname**
command is very fast
so everyone's job
should finish in a few
seconds.



When it is finished you will
have a new file named
slurm-{your job id}.out.

```
[vanilla@hopper intro_workshop]$ ls  
code data pbs slurm slurm-5252.out
```



When it is finished you will have a new file named slurm-{your job id}.out.

```
[vanilla@hopper intro_workshop]$ cat slurm-5252.out  
What do you see?
```

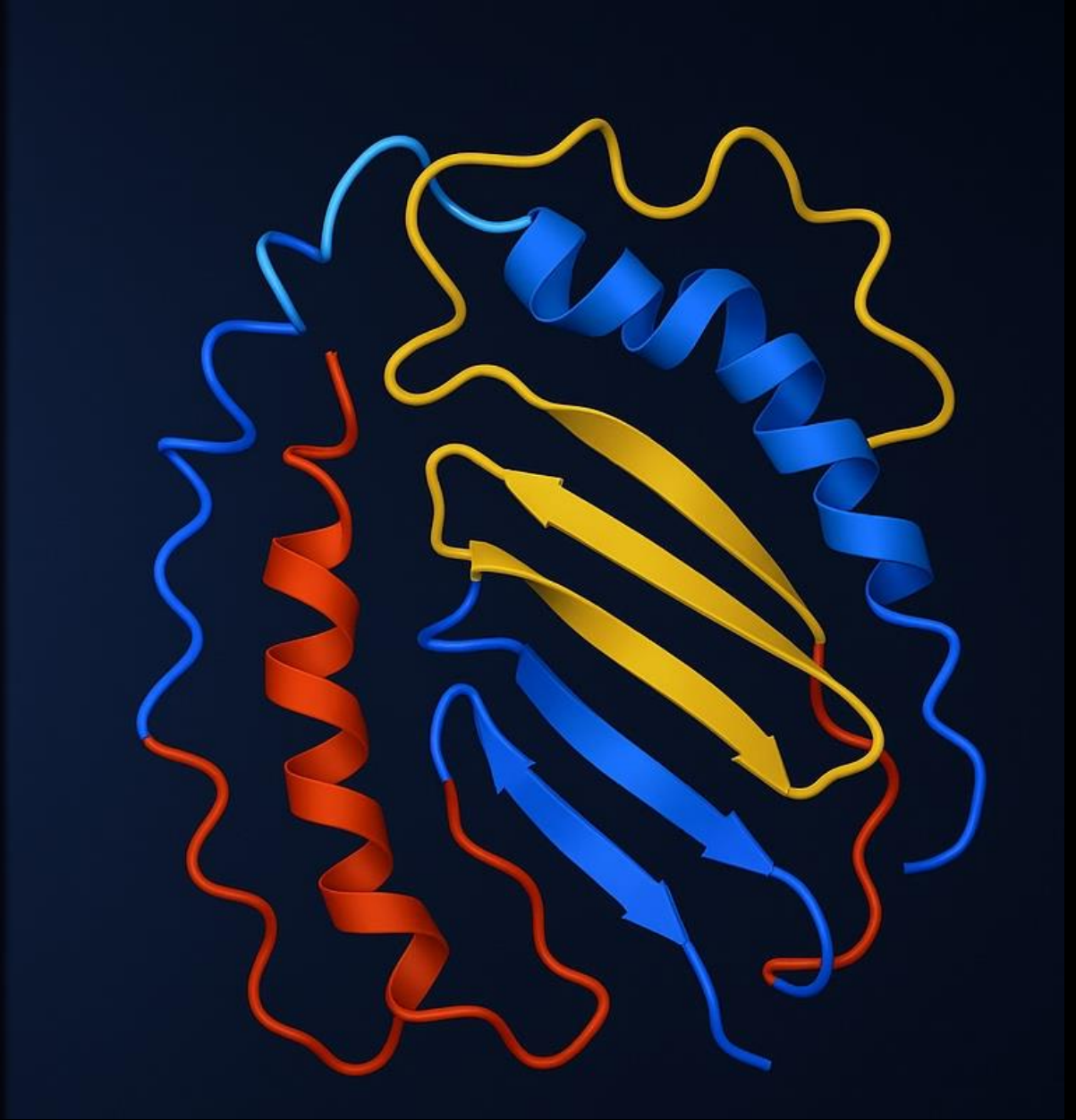
```
[vanilla@hopper intro_workshop]$ ls  
code data pbs slurm slurm-5252.out
```



When it is finished you will have a new file named `slurm-{your job id}.out`.

```
[vanilla@hopper intro_workshop]$ cat slurm-5252.out  
To request GPUs, add --gpus-per-node X or --gpus X, where X is the desired number of GPUs.  
Job 3687623 running on hopper011  
hopper011  
hopper011  
hopper011  
hopper011
```

Crystallography



Crystallography

Our goal...

X-ray diffraction pattern of spots scattered by the periodic structure of molecules in a crystal

Computationally Inferred molecular structure from the pattern of spots.



Collaborative Computational Project Number 4



A collection of programs designed to help crystallographers determine the 3D structures of biological macromolecules (proteins, DNA, etc.) from X-ray diffraction data.

It's been under active development since the 1970s, primarily in the UK, with global contributions from structural biology labs.

Collaborative Computational Project Number 4



```
vanilla@hopper:~$ cd ~/workshops/crystallography
```

```
vanilla@hopper:crystallography$ tree
```

```
.
├── data
│   └── thaumatin.pdb
└── slurm
    ├── integrate_xray_images.slurm
    └── solve_structure.slurm
```

Collaborative Computational Project Number 4



```
vanilla@hopper:~$ module spider ccp4
```

```
-----  
ccp4: ccp4/9.0.006  
-----
```

This module can be loaded directly: `module load ccp4/9.0.006`

Help:

an interactive molecular graphics program for macromolecular model building, validation, and visualization, primarily used in crystallography and cryo-EM.

Collaborative Computational Project Number 4



```
vanilla@hopper:~$ module spider ccp4
```

```
-----  
ccp4: ccp4/9.0.006  
-----
```

```
This module can b
```

```
Help:
```

```
  an interactive molecular graphics program for macromolecular model building, validation, and visualization, primarily used  
in crystallography and cryo-EM.
```



**Search for software by name
(if not installed ask us to install it)**

Collaborative Computational Project Number 4



```
vanilla@hopper:~$ module spider ccp4
```

```
-----  
ccp4: ccp4/9.0.006  
-----
```

This module can be loaded directly: `module load ccp4/9.0.006`

**Version of the software Sumaya
installed for you**

H... validation, and visualization, primarily used
in c...

Collaborative Computational Project Number 4



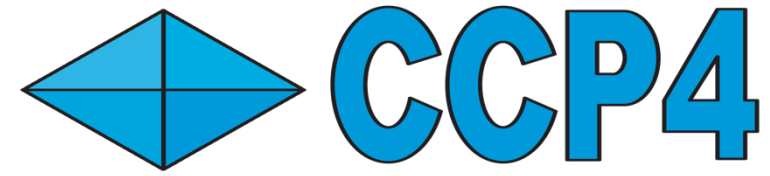
```
vanilla@hopper:~$ module load ccp4
```

```
Executing ccp4 environment script: /opt/local/ccp4/ccp4-9/bin/ccp4.setup-sh
```

```
Environment variables set by this script persist after the module is unloaded.
```

```
You are on the hopper head node. Be sure to run ccp4 tools on a compute node, e.g. "srun --x11 --pty coot"
```

Collaborative Computational Project Number 4

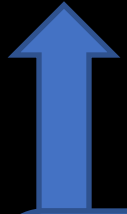


```
vanilla@hopper:~$ module load ccp4
```

```
Executing ccp4 environment script: /opt/local/ccp4/ccp4-9/bin/ccp4.setup-sh
```

```
Environment variables set by this script persist after the module is unloaded.
```

```
You are on the hopper head node. Be sure to run ccp4 tools on a compute node, e.g. "srun --x11 --pty coot"
```



**Reminder to run these tools on the
compute nodes with srun or sbatch**

Collaborative Computational Project Number 4



```
vanilla@hopper:~$ module load ccp4
```

```
Executing ccp4 environment script: /opt/local/ccp4/ccp4-9/bin/ccp4.setup-sh
```

```
Environment variables set by this script persist after the module is unloaded.
```

```
You are on the hopper head node. Be sure to run ccp4 tools on a compute node, e.g. "srun --x11 --pty coot"
```



--x11 allows us to run graphical programs like coot.
See the video tutorial on X11 forwarding in [in this playlist](#) for more information.

Image set for Thaumatin

```
vanilla@hopper: crystallography $ ls -lh  
/projects/shared/workshops/crystallography/vmxi_thaumatin/  
total 514M  
-rw-r--r-- 1 mfricke users 512M Mar 25 22:13 image_15799_data_000001.h5  
-rw-r--r-- 1 mfricke users  44K Mar 25 22:07 image_15799_master.h5  
-rw-r--r-- 1 mfricke users 119K Mar 25 22:07 image_15799_meta.hdf5  
-rw-r--r-- 1 mfricke users  49K Mar 25 22:07 image_15799.nxs
```

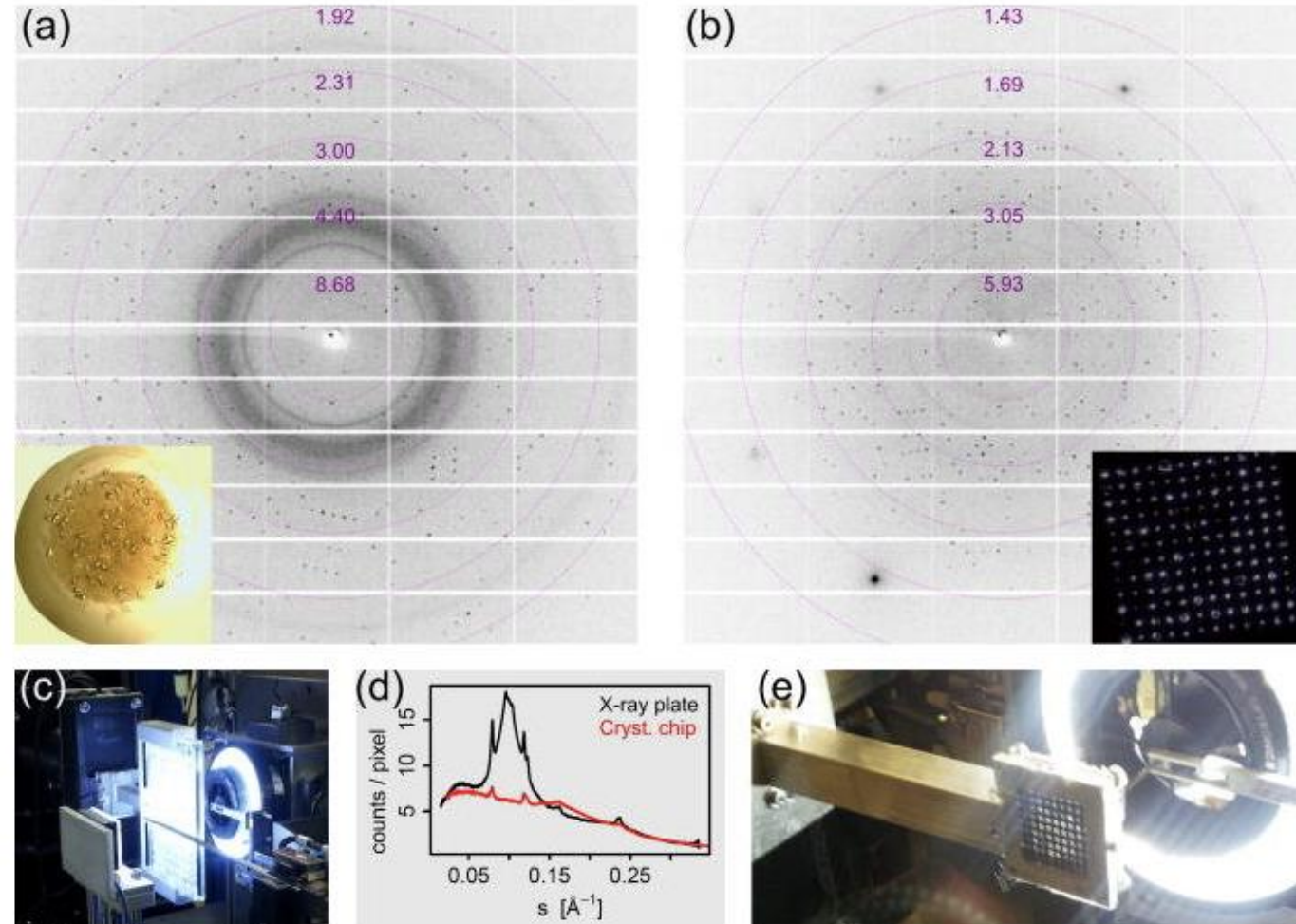
Katemfe fruit (*Thaumatococcus daniellii*), a tropical West African plant known for producing **thaumatin**—a protein that's thousands of times sweeter than sugar.



X-ray diffraction pattern of Crystallised Thaumatin



Mueller, C. & Marx, Alexander & Epp, S. & Zhong, Yinpeng & Kuo, A. & Balo, Aidin & Soman, J. & Schotte, F. & Lemke, H. & Owen, R. & Pai, Emil & Pearson, Arwen & Olson, John & Anfinrud, Philip & Ernst, O. & Miller, R.. (2015). Fixed target matrix for femtosecond time-resolved and in situ serial micro-crystallography. Structural Dynamics. 2. 10.1063/1.4928706.





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CXIDB ID 49

Deposition Summary

Depositor:	Michihiro Sugahara and Takanori Nakane
Contact:	msu...@bs.s.u-tokyo.ac.jp
Deposition date:	2017-04-07
Last modified:	2017-04-07
DOI:	10.11577/1350030
Associated PDB:	5WR8 , 5WRC , 5WR9 , 5WRA , 5WRB

Publication Details

Title:	Hydroxyethyl cellulose matrix applied to serial crystallography
Authors:	Michihiro Sugahara et al.
Journal:	Scientific Reports
Year:	2017
DOI:	10.1038/s41598-017-00761-0

Experimental Conditions

Method:	Serial Femtosecond Crystallography
Sample:	Thaumatococcus
Wavelength:	1.24 Å (10 keV)
Lightsource:	SACLA
Beamline:	BL3

Description

Please check the README file inside the metadata directory for more information about the dataset.



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Please give proper credit via citations according to established scientific practice.

X-ray Databank

Image set for Thaumatin

```
vanilla@hopper: crystallography $ ls -lh  
/projects/shared/workshops/crystallography/vmxi_thaumatin/  
total 514M  
-rw-r--r-- 1 mfricke users 512M Mar 25 22:13 image_15799_data_000001.h5  
-rw-r--r-- 1 mfricke users  44K Mar 25 22:07 image_15799_master.h5  
-rw-r--r-- 1 mfricke users 119K Mar 25 22:07 image_15799_meta.hdf5  
-rw-r--r-- 1 mfricke users  49K Mar 25 22:07 image_15799.nxs
```

(PS I used this command to get the data, so you don't have to)

```
dials.data get -q vmxi_thaumatin
```

```
Downloading https://zenodo.org/record/2547566/files/image_15799.nxs
```

```
Downloading https://zenodo.org/record/2547566/files/image_15799_data_000001.h5
```

```
Downloading https://zenodo.org/record/2547566/files/image_15799_master.h5
```

```
Downloading https://zenodo.org/record/2547566/files/image_15799_meta.hdf
```

DIALS (Diffraction Integration for Advanced Light Sources)

Integration of Thaumatin Images

```
vanilla@hopper: crystallography $ cat slurm/integrate_xray_images.slurm
```

```
#!/bin/bash

#SBATCH --job-name xray_int
#SBATCH --partition debug
#SBATCH --output xray_int_%j.out
#SBATCH --error xray_int_%j.out
#SBATCH --time 0:05:00
#SBATCH --nodes 1
#SBATCH --ntasks-per-node 1
#SBATCH --cpus-per-task 8
#SBATCH --mem 8G
#SBATCH --mail-type ALL
#SBATCH --mail-user your.email@unm.edu

# Load CCP4 and configure environment
module load ccp4
export OMP_NUM_THREADS=$SLURM_CPUS_PER_TASK
cd $SLURM_SUBMIT_DIR

# Define input dataset path
INPUT_PATH=/projects/shared/workshops/crystallography/vmxi_thaumatin/image_15799_master.h5
OUTPUT_PATH=~/.workshops/crystallography/data/integration_output

# Create output directory
mkdir -p $OUTPUT_PATH
cd $OUTPUT_PATH

# Run xia2 with input and working directory
xia2 pipeline=dials \
  image=$INPUT_PATH \
  multiprocessing.mode=serial \
  multiprocessing.nproc=$SLURM_CPUS_PER_TASK \
  trust_beam_centre=False \
  read_all_image_headers=False
```

Integration of Thaumatin Images

```
vanilla@hopper: crystallography $ cat slurm/integrate_xray_images.slurm
```

```
#!/bin/bash
```

```
#SBATCH --job-name xray_int
```

```
#SBATCH --partition debug
```

```
#SBATCH --output xray_int_%j.out
```

```
#SBATCH --error xray_int_%j.out
```

```
#SBATCH --time 0:05:00
```

```
#SBATCH --nodes 1
```

```
#SBATCH --ntasks-per-node 1
```

```
#SBATCH --cpus-per-task 8
```

```
#SBATCH --mem 8G
```

```
#SBATCH --mail-type ALL
```

```
#SBATCH --mail-user your.email@unm.edu
```

Integration of Thaumatin Images

```
vanilla@hopper: crystallography $ cat slurm/integrate_xray_images.slurm
```

```
...
```

```
# Load CCP4 and configure environment
```

```
module load ccp4
```

```
export OMP_NUM_THREADS=$SLURM_CPUS_PER_TASK
```

```
cd $SLURM_SUBMIT_DIR
```

Integration of Thaumatin Images

```
vanilla@hopper: crystallography $ cat slurm/integrate_xray_images.slurm
```

```
...
```

```
# Define input dataset path
```

```
INPUT_PATH=/projects/shared/workshops/crystallography/vmxi_thaumatin/image_15799_master.h5
```

```
OUTPUT_PATH=~/.workshops/crystallography/data/integration_output
```

```
# Create output directory
```

```
mkdir -p $OUTPUT_PATH
```

```
cd $OUTPUT_PATH
```

Integration of Thaumatin Images

```
vanilla@hopper: crystallography $ cat slurm/integrate_xray_images.slurm
```

```
...
```

```
# Run xia2 with input and working directory
```

```
xia2 pipeline=dials \  
  image=$INPUT_PATH \  
  multiprocessing.mode=serial \  
  multiprocessing.nproc=$SLURM_CPUS_PER_TASK \  
  trust_beam_centre=False \  
  read_all_image_headers=False
```

xia2 is data reduction pipeline that wraps around **DIALS**

Integration of Thaumatin Images

```
vanilla@hopper: crystallography $ cat slurm/integrate_xray_images.slurm
```

```
...
```

```
# Run xia2 with input and working directory
```

```
xia2 pipeline=dials \
```

```
image=$INPUT_PATH \
```

```
multiprocessing.mode=serial \
```

```
multiprocessing.nproc=$SLURM_CPUS_PER_TASK \
```

```
trust_beam_centre=False \
```

```
read_all_image_headers=False
```

This BASH variable contains the number of CPUs Slurm allocated

Integration of Thaumatin Images

```
vanilla@hopper: crystallography $ sbatch slurm/integrate_xray_images.slurm  
sbatch: Account not specified in script or ~/.default_slurm_account, using latest project  
Submitted batch job 3687646
```

```
vanilla@hopper: crystallography $ watch squeue --me
```

- 1) Submit the batch job to the Slurm scheduler
- 2) Monitor the progress of your job with `squeue --me`

Integration of Thaumatin Images

Every 2.0s: squeue --me hopper: Wed Mar 26 10:49:35 2025

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
3687646	debug	xray_int	vanilla	R	1:24	1	hopper011

Check the job status every 2 seconds

Once your job is in the R state check the name of the node on which it's running

Integration of Thaumatin Images

Every 2.0s: squeue --me hopper: Wed Mar 26 10:49:35 2025

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
3687646	debug	xray_int	vanilla	R	1:24	1	hopper011

Check the job status every 2 seconds

Once your job is in the R state check the name of the node on which it's running

vanilla@hopper: crystallography \$ ssh hopper011

vanilla@hopper011: crystallography \$ htop

1[	0.0%]	9[	0.0%]	17[93.4%]	25[96.6%]
2[	0.0%]	10[	1.3%]	18[	1.3%]
3[	0.0%]	11[	0.0%]	19[92.8%]	27[94.0%]
4[	2.0%]	12[	0.0%]	20[	0.0%]
5[	0.0%]	13[	0.0%]	21[96.7%]	29[98.7%]
6[	0.0%]	14[	0.0%]	22[	0.0%]
7[	0.7%]	15[	0.0%]	23[98.0%]	31[96.7%]
8[	0.0%]	16[	0.0%]	24[	0.0%]
Mem[			14.2G/92.9G]	Tasks: 50, 117 thr, 443 kthr;	13 running
Swp[]			996K/85.2G]	Load average: 5.53 1.78 0.82	
				Uptime: 34 days, 21:39:21	

PID	USER	PRI	NI	VIRT	RES	SHR	S	CPU%	MEM%	TIME+	Command
481822	mfricke	20	0	990M	356M	44296	R	105.2	0.4	0:02.78	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481795	mfricke	20	0	964M	331M	44168	R	101.9	0.3	0:02.71	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481823	mfricke	20	0	1003M	369M	44296	R	97.3	0.4	0:02.64	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481797	mfricke	20	0	990M	356M	44292	R	93.3	0.4	0:02.66	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481806	mfricke	20	0	991M	358M	44296	R	92.7	0.4	0:02.44	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481796	mfricke	20	0	990M	356M	44228	R	92.0	0.4	0:02.68	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481814	mfricke	20	0	990M	356M	44296	R	90.0	0.4	0:02.49	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481798	mfricke	20	0	990M	356M	44292	R	87.4	0.4	0:02.50	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481623	mfricke	20	0	35964	4952	3488	R	1.3	0.0	0:00.33	htop
481713	mfricke	20	0	442M	330M	101M	S	0.7	0.3	0:06.67	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
1	root	20	0	232M	11612	8660	S	0.0	0.0	0:19.65	/sbin/init
534	root	20	0	128M	43856	40204	S	0.0	0.0	0:08.46	/usr/lib/systemd/systemd-journald
581	root	20	0	98124	9308	6976	S	0.0	0.0	0:02.25	/usr/lib/systemd/systemd-udev
756	rpc	20	0	67328	5580	4848	S	0.0	0.0	0:05.97	/usr/bin/rpcbind -w -f
780	root	20	0	702M	17976	5912	S	0.0	0.0	1:10.33	/warewulf/wwclient
788	root	20	0	236M	9100	7828	S	0.0	0.0	0:32.04	/usr/sbin/rsyslogd -n
790	root	20	0	109M	3920	3256	S	0.0	0.0	0:00.00	/usr/sbin/gssproxy -D

F1HelpF2SetupF3SearchF4FilterF5TreeF6SortByF7Nice -F8Nice +F9KillF10Quit

vanilla@hopper011: crystallography \$ htop

1[0.0%] 9[0.0%] 17[93.4%] 25[96.6%]
2[0.0%] 10[1.3%] 18[1.3%] 26[0.7%]
3[0.0%] 11[0.0%] 19[92.8%] 27[94.0%]
4[2.0%] 12[0.0%] 20[0.0%] 28[0.0%]
5[0.0%] 13[0.0%] 21[96.7%] 29[98.7%]
6[0.0%] 14[0.0%] 22[0.0%] 30[3.9%]
7[0.7%] 15[0.0%] 23[98.0%] 31[96.7%]
8[0.0%] 16[0.0%] 24[0.0%] 32[0.0%]
Mem[|||||] 14.2G/96G Tasks: 50, 117 thr, 443 kthr; 13 running
Swp[||] 996K 1G Load average: 5.53 1.78 0.82
Uptime: 34 days, 21:39:21

PID	USER	PRI	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	Command
481828	root	20	0	228M	250M	11200K	S	95.2	0.4	0:02.78	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481829	root	20	0	228M	250M	11200K	S	9.9	0.3	0:02.71	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481830	root	20	0	228M	250M	11200K	S	0.3	0.4	0:02.64	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481831	root	20	0	228M	250M	11200K	S	0.3	0.4	0:02.66	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481832	root	20	0	228M	250M	11200K	S	0.0	0.4	0:02.44	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481833	root	20	0	228M	250M	11200K	S	0.0	0.4	0:02.68	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481834	root	20	0	228M	250M	11200K	S	0.0	0.4	0:02.49	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481835	root	20	0	228M	250M	11200K	S	0.4	0.4	0:02.50	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481836	root	20	0	228M	250M	11200K	S	0.3	0.0	0:00.33	htop
481837	root	20	0	228M	250M	11200K	S	0.7	0.3	0:06.67	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481838	root	20	0	228M	250M	11200K	S	0.0	0.0	0:19.65	/sbin/init
534	root	20	0	128M	43856	40204	S	0.0	0.0	0:08.46	/usr/lib/systemd/systemd-journald
581	root	20	0	98124	9308	6976	S	0.0	0.0	0:02.25	/usr/lib/systemd/systemd-udevd
756	rpc	20	0	67328	5580	4848	S	0.0	0.0	0:05.97	/usr/bin/rpcbind -w -f
780	root	20	0	702M	17976	5912	S	0.0	0.0	1:10.33	/warewulf/wwclient
788	root	20	0	236M	9100	7828	S	0.0	0.0	0:32.04	/usr/sbin/rsyslogd -n
790	root	20	0	109M	3920	3256	S	0.0	0.0	0:00.00	/usr/sbin/gssproxy -D

F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice -F8Nice +F9Kill F10Quit

Here are the 8 copies of our program running on 8 CPUs

vanilla@hopper011: crystallography \$ htop

1[	0.0%	9[	0.0%	17[	93.4%	25[	96.6%
2[	0.0%	10[	1.3%	18[	1.3%	26[	0.7%
3[	0.0%	11[	0.0%	19[	92.8%	27[	94.0%
4[	2.0%	12[	0.0%	20[	0.0%	28[	0.0%
5[	0.0%	13[	0.0%	21[	96.7%	29[	98.7%
6[	0.0%	14[	0.0%	22[	0.0%	30[	3.9%
7[	0.7%	15[	0.0%	23[	98.0%	31[	96.7%
8[	0.0%	16[	0.0%	24[	0.0%	32[	0.0%
Mem[			14.2G/92.9G	Tasks: 50, 117 thr, 443 kthr; 13 running			
Swp[			996K/85.2G	Load average: 5.53 1.78 0.82			
Uptime: 34 days, 21:39:21							

PID	USER	PRI	NI	VIRT	RES	SHR	S	CPU%	MEM%	TIME+	Command
481822	root	20	0	232M	352M	11228	R	95.2	0.4	0:02.78	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481823	root	20	0	232M	352M	11228	R	9.9	0.3	0:02.71	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481824	root	20	0	232M	352M	11228	R	0.3	0.4	0:02.64	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481825	root	20	0	232M	352M	11228	R	0.3	0.4	0:02.66	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481826	root	20	0	232M	352M	11228	R	0.7	0.4	0:02.44	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481827	root	20	0	232M	352M	11228	R	0.0	0.4	0:02.68	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481828	root	20	0	232M	352M	11228	R	0.0	0.4	0:02.49	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481829	root	20	0	232M	352M	11228	R	0.4	0.4	0:02.50	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
481830	root	20	0	232M	352M	11228	R	0.3	0.0	0:00.33	htop
481831	root	20	0	232M	352M	11228	R	0.7	0.3	0:06.67	/opt/local/ccp4/ccp4-9/libexec/python3 /opt/local/ccp4/
534	root	20	0	128M	43856	40204	S	0.0	0.0	0:19.65	/sbin/init
581	root	20	0	98124	9308	6976	S	0.0	0.0	0:08.46	/usr/lib/systemd/systemd-journald
756	rpc	20	0	67328	5580	4848	S	0.0	0.0	0:02.25	/usr/lib/systemd/systemd-udevd
780	root	20	0	702M	17976	5912	S	0.0	0.0	0:05.97	/usr/bin/rpcbind -w -f
788	root	20	0	236M	9100	7828	S	0.0	0.0	0:10.33	/warewulf/wwclient
790	root	20	0	109M	3920	3256	S	0.0	0.0	0:32.04	/usr/sbin/rsyslogd -n
								0.0	0.0	0:00.00	/usr/sbin/gssproxy -D

vanilla@hopper011: crystallography \$ exit

Enter exit to log out of the compute node.

Be careful to not log out of the head node.

**If your prompt shows hopper
don't type exit**

```
vanilla@hopper: crystallography $ watch queue --me
```

Check on our job status again



Integration of Thaumatin Images

Every 2.0s: squeue --me hopper: Wed Mar 26 10:49:35 2025

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
3687646	debug	xray_int	vanilla	R	3:24	1	hopper011

Integration of Thaumatin Images

Every 2.0s: squeue --me hopper: Wed Mar 26 10:49:35 2025

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
-------	-----------	------	------	----	------	-------	------------------

When your job finishes exit watch with *ctrl-c* or 'q'

Integration of Thaumatin Images

```
vanilla@hopper: crystallography $ ls  
data slurm xray_int_{your job ID}.out
```

You will have a new slurm output file.

Integration of Thaumatin Images

```
vanilla@hopper: crystallography $ cat xray_int_{your job ID}.out
```

Unit cell (with estimated std devs):

58.6134(2) 58.6134(2) 151.3821(8)

90.0 90.0 90.0

mtz_unmerged format:

Scaled reflections (NATIVE): /users/vanilla/workshops/crystallography/data/integration_output/DataFiles/AUTOMATIC_DEFAULT_scaled_unmerged.mtz

mtz format:

Scaled reflections: /users/vanilla/workshops/crystallography/data/integration_output/DataFiles/AUTOMATIC_DEFAULT_free.mtz

Processing took 00h 02m 59s

XIA2 used... ccp4, dials, dials.scale, xia2

Here are the appropriate citations (BIBTeX in xia2-citations.bib.)

Beilsten-Edmands, J. et al. (2020) Acta Cryst. D76, 385-399.

Winn, M. D. et al. (2011) Acta Cryst. D67, 235-242.

Winter, G. (2010) J. Appl. Cryst. 43, 186-190.

Winter, G. et al. (2018) Acta Cryst. D74, 85-97.

Status: normal termination

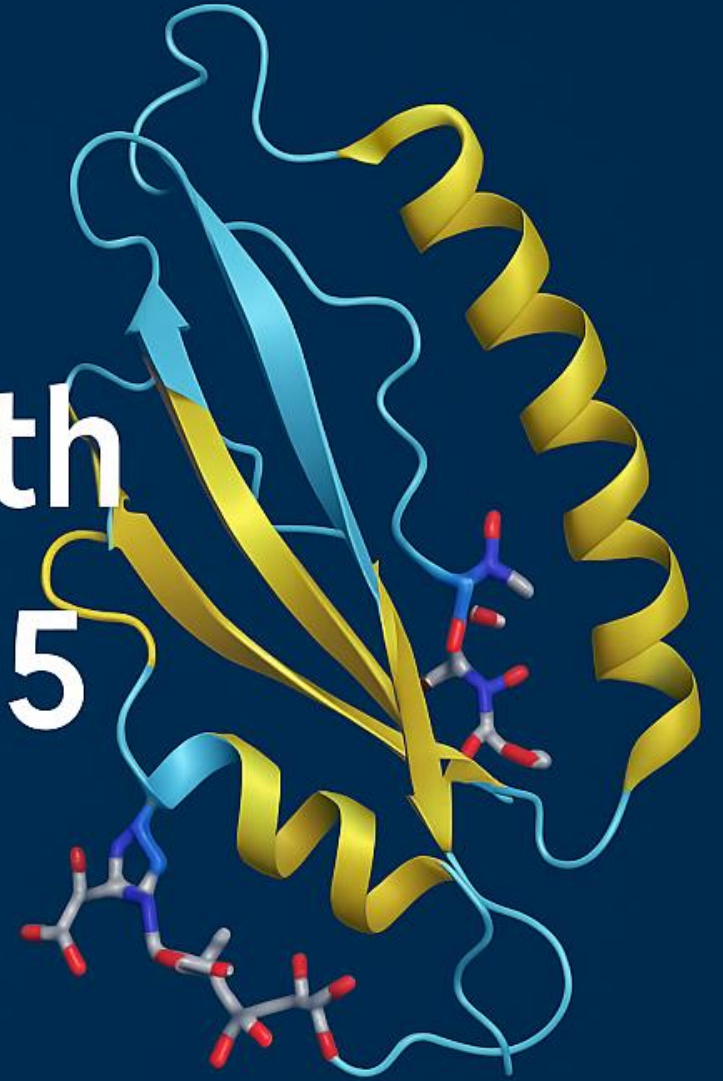
Integration of Thaumatin Images

vanilla@hopper: crystallography \$ tree data

```
data
├── integration_output
│   ├── automatic.xinfo
│   └── DataFiles
│       ├── AUTOMATIC_DEFAULT_free.mtz
│       ├── AUTOMATIC_DEFAULT_NATIVE_SWEEP1.expt
│       ├── AUTOMATIC_DEFAULT_NATIVE_SWEEP1_INTEGRATE.mtz
│       ├── AUTOMATIC_DEFAULT_NATIVE_SWEEP1.refl
│       ├── AUTOMATIC_DEFAULT_scaled.expt
│       ├── AUTOMATIC_DEFAULT_scaled.mtz
│       ├── AUTOMATIC_DEFAULT_scaled.refl
│       ├── AUTOMATIC_DEFAULT_scaled.sca
│       ├── AUTOMATIC_DEFAULT_scaled_unmerged.mtz
│       ├── AUTOMATIC_DEFAULT_scaled_unmerged.sca
│       ├── xia2.cif
│       └── xia2.mmcif.bz2
├── DEFAULT
│   └── NATIVE
│       └── SWEEP1
```

12 directories, 154 files

Structure Solving with MOLREP & REFMAC5



molrep

```
+-----+  
|  
|      --- MOLREP ---  
|    /Vers 11.9.02; 28.02.2022/  
|  
+-----+
```

```
molrep HKLIN data.mtz MODEL model.pdb  
labin F=FP SIGF=SIGFP  
mode MR_AUTO  
nmon 1  
doc y  
_end
```

Use experimental data and model to place the molecule in the unit cell.

You can run this interactively and enter the commands you want to run...

Solving for Thaumatin Structure

```
molrep \  
  HKLIN $INT_PATH/DataFiles/AUTOMATIC_DEFAULT_free.mtz \  
  MODEL $PDB_PATH << EOF  
labin F=FP SIGF=SIGFP  
mode MR_AUTO  
nmon 1  
doc y  
_end  
EOF
```

... but we will use the batch system not interactive sessions. So, we will send the commands to molrep in a script by wrapping them in an EOF (same with refmac5)

refmac5

```
#####  
#####  
#####  
### CCP4 9.0.006: Refmac      version 5.8.0430 : 05/20/24##  
#####  
User: vanilla  Run date: 26/ 3/2025 Run time: 11:41:25
```

REFMAC5 refines atomic models in crystallography, improving fit to X-ray data while preserving chemical geometry.

Solving for Thaumatin Structure

```
vanilla@hopper: crystallography $ less slurm/solve_structure.slurm
```

```
#!/bin/bash
#SBATCH --job-name solve_structure
#SBATCH --partition debug
#SBATCH --output solve_structure_%j.out
#SBATCH --error solve_structure_%j.out
#SBATCH --time 0:05:00
#SBATCH --nodes 1
#SBATCH --ntasks-per-node 1
#SBATCH --cpus-per-task 8
#SBATCH --mem 8G
#SBATCH --mail-type ALL
#SBATCH --mail-user your.email@university.edu
# Load CCP4 environment
module load ccp4
export OMP_NUM_THREADS=$SLURM_CPUS_PER_TASK
cd $SLURM_SUBMIT_DIR
INPUT_PATH=~/.workshops/crystallography/data
OUTPUT_PATH=$INPUT_PATH/solver_output

# Path to the output of the image integrator (xia2)
INT_PATH=$INPUT_PATH/integration output
```

Solving for Thaumatin Structure

```
# Create output directory for this job
```

```
mkdir -p $OUTPUT_PATH
```

```
cd $OUTPUT_PATH
```

```
# Molecular replacement using molrep
```

```
molrep \
```

```
  HKLIN $INT_PATH/DataFiles/AUTOMATIC_DEFAULT_free.mtz \
```

```
  MODEL $PDB_PATH << EOF
```

```
labin F=FP SIGF=SIGFP
```

```
mode MR_AUTO
```

```
nmon 1
```

```
doc y
```

```
_end
```

```
EOF
```

```
# Refinement using refmac5
```

```
refmac5 \
```

```
  hklin $INT_PATH/DataFiles/AUTOMATIC_DEFAULT_free.mtz \
```

```
  hklout $OUTPUT_PATH/refined.mtz \
```

```
  xyzin $OUTPUT_PATH/molrep.pdb \
```

```
  xyzout $OUTPUT_PATH/refined.pdb << EOF >
```

```
$OUTPUT_PATH/refmac.log
```

```
make hydrogen ALL
```

```
ncyc 10
```

```
weight AUTO
```

```
monitor MEDIUM
```

```
scales PEAK
```

```
labout FP=F SIGFP=SIGF FREE=FreeR_flag
```

```
end
```

```
EOF
```

Solving for Thaumatin Structure

```
# Create output directory for this job
```

```
mkdir -p $OUTPUT_PATH
```

```
cd $OUTPUT_PATH
```

```
# Molecular replacement using molrep
```

```
molrep \
```

```
  HKLIN $INT_PATH/DataFiles/AUTOMATIC_DEFAULT_free.mtz \
```

```
  MODEL $PDB_PATH << EOF
```

```
labin F=FP SIGF=SIGFP
```

```
mode MR_AUTO
```

```
nmon 1
```

```
doc y
```

```
_end
```

```
EOF
```

This PDB gives us a starting point for the refinement

Solving for Thaumatin Structure

```
# Refinement using refmac5
refmac5 \
  hklin $INT_PATH/DataFiles/AUTOMATIC_DEFAULT_free.mtz \
  hklout $OUTPUT_PATH/refined.mtz \
  xyzin $OUTPUT_PATH/molrep.pdb \
  xyzout $OUTPUT_PATH/refined.pdb << EOF > $OUTPUT_PATH/refmac.log
make hydrogen ALL
ncyc 10
weight AUTO
monitor MEDIUM
scales PEAK
labout FP=F SIGFP=SIGF FREE=FreeR_flag
end
EOF
```

Solving for Thaumatin Structure

```
vanilla@hopper: crystallography $ sbatch slurm/solve_structure.slurm
```

```
vanilla@hopper: crystallography $ watch squeue --me
```

Every 2.0s: squeue --me

hopper: Wed Mar 26 11:54:15 2025

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
3687680	debug	solve_st	vanilla	R	1:29	1	hopper011

Solving for Thaumatin Structure

```
vanilla@hopper:crystallography $ ls
data slurm solve_structure_3687680.out xray_int_3687646.out

vanilla@hopper:crystallography $ cat solve_structure_3687680.out
```

```
--- Summary (V2) ---
+-----+
| RF TF theta phi chi tx ty tz TF/sg Tcoef Score |
+-----+
| 1 1 1 0.00 0.00 0.09 0.212 0.413 0.210 30.04 1.000 0.72199|
| 2 13 1 11.43 88.85 155.77 0.330 0.425 0.339 3.68 1.000 0.29457|
| 3 12 2 13.34 91.75 134.42 0.010 0.301 0.066 3.26 1.000 0.29332|
| 4 4 2 23.97 87.94 129.70 0.407 0.112 0.230 3.39 1.000 0.28992|
| 5 14 2 8.49 70.24 140.71 0.378 0.224 0.120 3.11 1.000 0.28702|
| 6 7 1 19.89 51.50 179.63 0.241 0.071 0.398 4.11 1.000 0.27902|
| 7 9 14 11.53 92.47 156.62 0.960 0.410 0.029 2.58 0.900 0.25990|
| 8 10 12 42.11 44.11 156.36 0.863 0.392 0.497 2.81 0.900 0.25988|
| 9 8 11 92.72 136.00 81.81 0.874 0.228 0.070 3.01 0.900 0.25872|
| 10 5 13 23.90 86.48 129.03 0.375 0.062 0.379 2.86 0.900 0.25821|
| 11 2 14 73.81 89.88 71.61 0.854 0.448 0.480 2.71 0.900 0.25792|
| 12 3 13 12.31 81.24 142.53 0.043 0.294 0.326 2.59 0.900 0.25447|
| 13 6 11 19.59 50.07 178.92 0.710 0.046 0.370 3.34 0.900 0.25107|
| 14 11 8 19.08 48.47 178.54 0.982 0.339 0.178 3.23 0.900 0.24901|
+-----+
```

Summary table showing top molecular replacement solutions ranked by their rotation and translation functions, with associated orientation angles (theta, phi, chi), shifts (tx, ty, tz), signal-to-noise (TF/ σ), and final placement scores.

Solving for Thaumatin Structure

```
vanilla@hopper:crystallography $ tree data
```

```
...
```

```
|— solver_output
|   |— molrep.btc
|   |— molrep.doc
|   |— molrep_mtz.cif
|   |— molrep.pdb
|   |— molrep.xml
|   |— refined.mmcif
|   |— refined.mtz
|   |— refined.pdb
|   |— refmac.log
|— thaumatin.pdb
```

Visualise the Thaumatin Structure

```
vanilla@hopper:crystallography $ exit
```

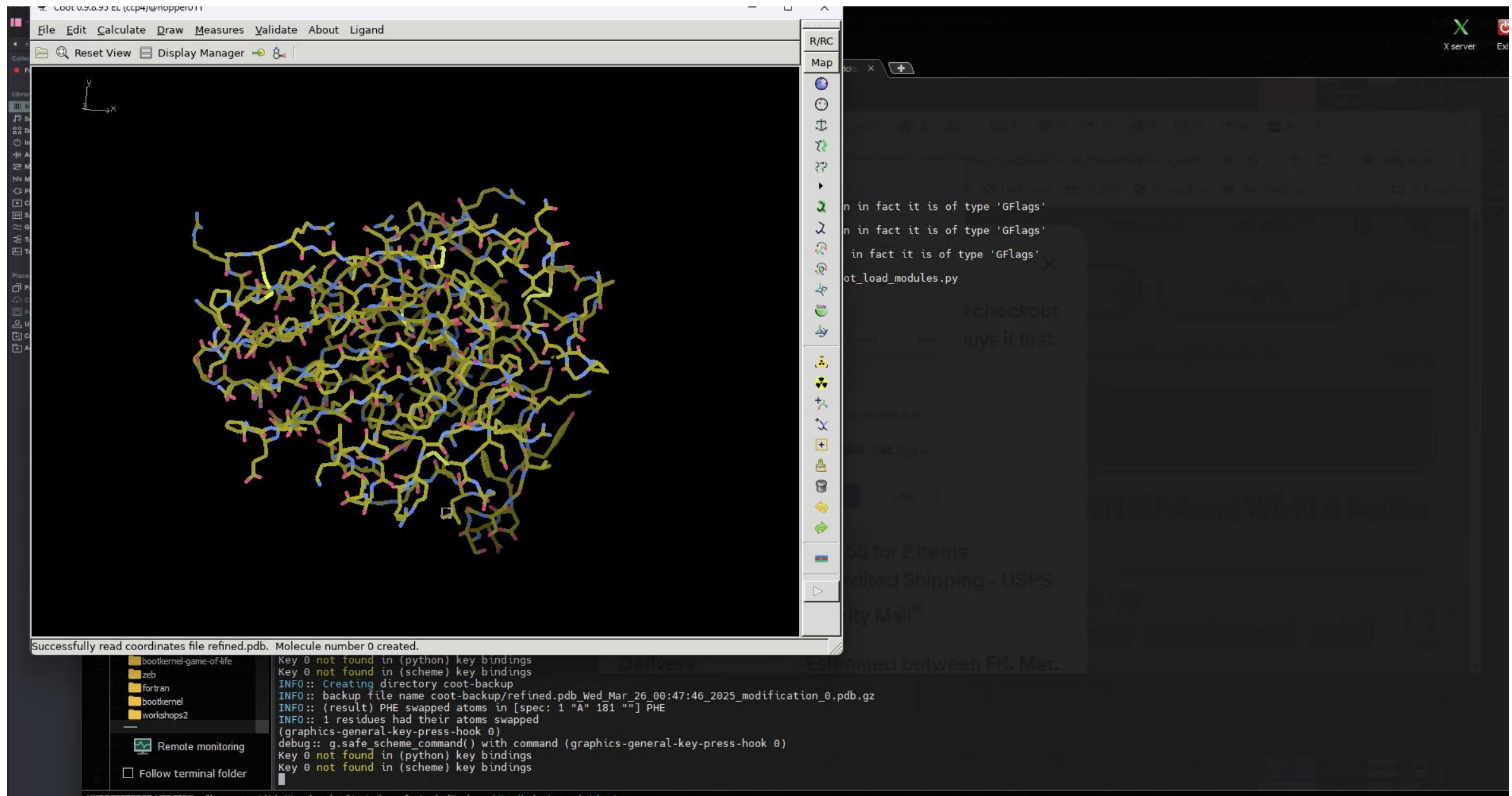
```
CARCWS-01:~ vanilla$ ssh -Y vanilla@hopper.alliance.unm.edu
```

-Y allows us to run graphical programs remotely.
See the video tutorial on X11 forwarding in [in this playlist](#) for more information.

Visualise the Thaumatin Structure

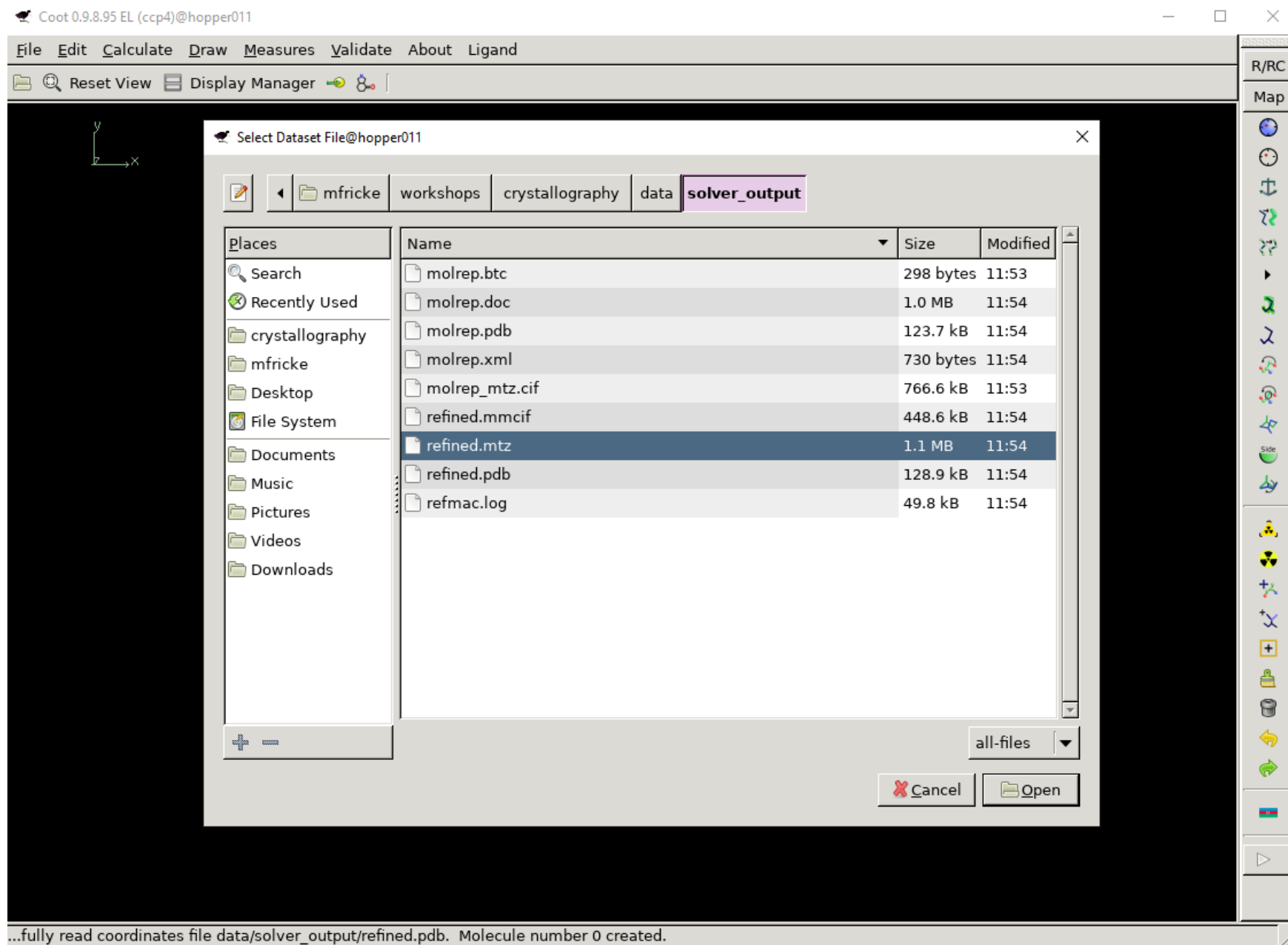
```
module load ccp4  
cd ~/workshops/crystallography  
srun --partition debug --x11 --pty coot --pdb data/solver_output/refined.pdb
```

--x11 allows us to run graphical programs remotely.
See the video tutorial on X11 forwarding in [in this playlist](#) for more information.

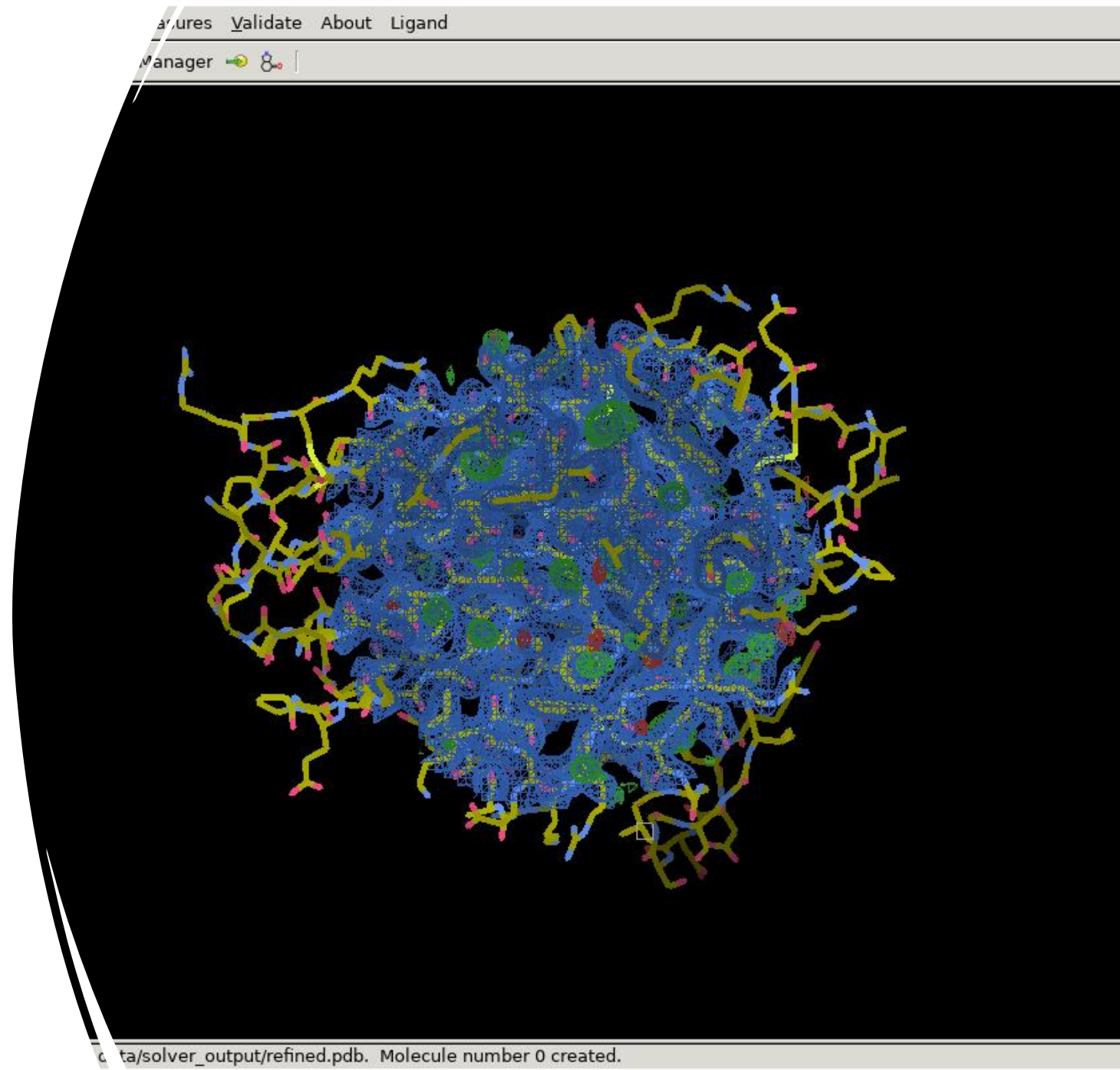


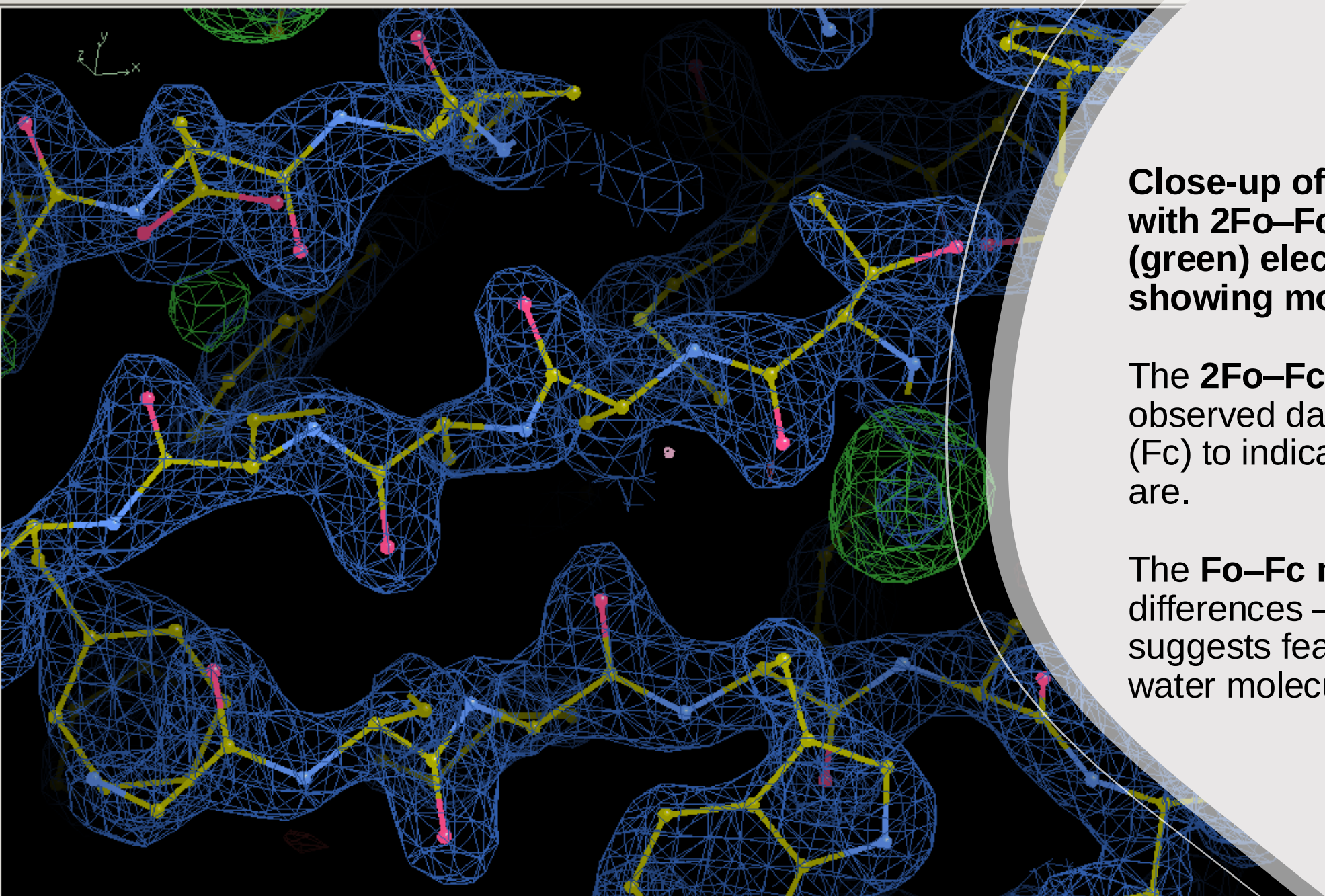
Refined thaumatin structure visualised in Coot following molecular replacement and refinement with MOLREP and REFMAC5.

Loading the refined structure and map in Coot: click 'File' → 'Auto Open MTZ' and select the appropriate .pdb and .mtz files from the solver_output directory.



*Refined thaumatin model
displayed with electron
density in Coot, showing
fit between atomic
coordinates and
experimental data.*





Close-up of the refined structure with **2Fo–Fc** (blue) and **Fo–Fc** (green) electron density maps, showing model fit.

The **2Fo–Fc map** combines observed data (Fo) and the model (Fc) to indicate where atoms likely are.

The **Fo–Fc map** highlights differences — green density suggests features like unmodelled water molecules.

Hardware Utilisation Performance

```
vanilla@hopper:crystallography $ ls  
data slurm solve_structure_3687680.out xray_int_3687646.out  
vanilla@hopper:crystallography $ seff 3687680
```

Job ID: 3687680

Cluster: hopper

User/Group: mfricke/users

State: COMPLETED (exit code 0)

Nodes: 1

Cores per node: 8

CPU Utilized: 00:00:42

CPU Efficiency: 12.50% of 00:05:36 core-walltime

Job Wall-clock time: 00:00:42

Memory Utilized: 207.62 MB

Memory Efficiency: 2.53% of 8.00 GB

How efficiently did our programs use the resources we allocated?

Not great. We should request fewer CPUs and less memory next time.

Hardware Utilisation Performance

```
vanilla@hopper:crystallography $ ls  
data slurm solve_structure_3687680.out xray_int_3687646.out  
vanilla@hopper:crystallography $ seff 3687646  
Job ID: 3687646  
Cluster: hopper  
User/Group: mfricke/users  
State: COMPLETED (exit code 0)  
Nodes: 1  
Cores per node: 8  
CPU Utilized: 00:08:34  
CPU Efficiency: 34.92% of 00:24:32 core-walltime  
Job Wall-clock time: 00:03:04  
Memory Utilized: 1.33 GB  
Memory Efficiency: 16.58% of 8.00 GB
```

How efficiently did our programs use the resources we allocated?
Better than the solve job, but we should still request fewer CPUs and less RAM.

Useful Slurm Commands

<code>squeue --me --long</code>	shows information about jobs you submitted
<code>squeue --me --start</code>	shows when slurm expects your job to start
<code>scancel jobid</code>	Cancels a job
<code>scancel --u \$USER</code>	Cancels all your jobs
<code>sacct</code>	Shows your job history
<code>seff jobid</code>	Shows how efficiently the hardware was used